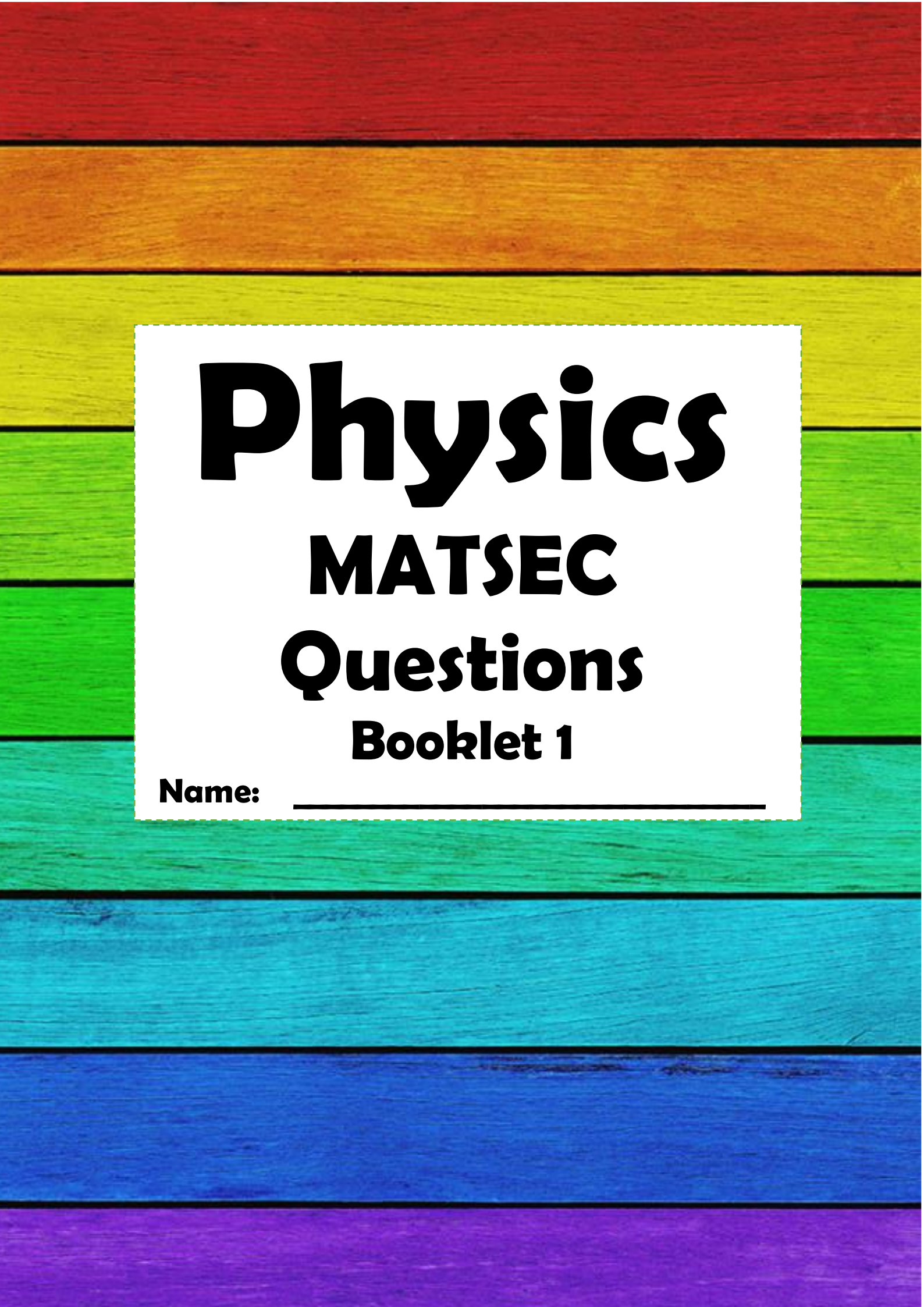




Physics

MATSEC

Questions

Booklet 1

Name: _____

You are requested to show your working and to write the units where necessary.
When necessary, take g , acceleration due to gravity, as 10m/s^2 .

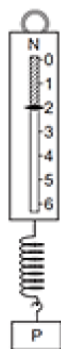
Density	$m = \rho V$		
Pressure	$F = p A$	$p = \rho g h$	
Moments	Moment = $F \times$ perpendicular distance		
Energy and Work	$PE = m g h$	$KE = \frac{1}{2} m v^2$	$W = F s$
	Work Done=energy converted		$E=P t$
Force and Motion	$m a =$ unbalanced force	$W = m g$	$v = u + a t$
	average speed = $\frac{\text{total distance}}{\text{total time}}$		$s = (u + v) \frac{t}{2}$
	$v^2= u^2 + 2 a s$	$s = u t + \frac{1}{2} a t^2$	momentum = $m v$
Waves	$\eta = \frac{\text{speed of light in air}}{\text{speed of light in medium}}$		$v = f \lambda$
	$\eta = \frac{\text{real depth}}{\text{apparent depth}}$	Magnification = $\frac{\text{image distance}}{\text{object distance}}$	
	Magnification = $\frac{\text{image height}}{\text{object height}}$		$T = \frac{1}{f}$
Electricity	$Q = I t$	$V = I R$	$E = Q V$
	$P = I V$	$R \propto \frac{L}{A}$	$E = I V t$
	$R_{\text{total}} = R_1 + R_2 + R_3$	$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2}$	
Electromagnetism	$\frac{V_p}{V_s} = \frac{N_p}{N_s}$	$V_p I_p = V_s I_s$	
Heat	$Q = m c \Delta \theta$		
Radioactivity	$A = Z + N$		
Other equations	Area of a triangle = $\frac{1}{2} b h$		Area of a trapezium = $\frac{1}{2} (a + b) h$
	Area of a circle = πr^2		

Forces

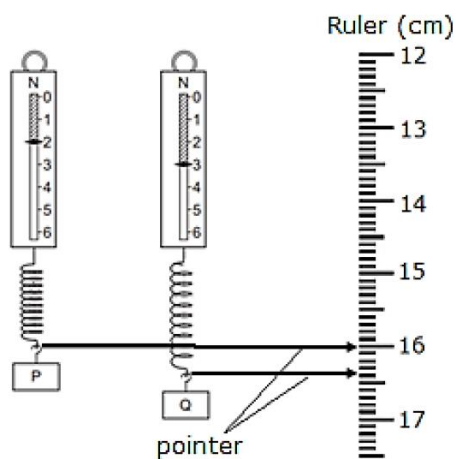
1. September 2020 Paper I

(10 marks)

The diagram below shows a Newton spring balance to which a metal block P has been attached.



- Copy the diagram name and label the **TWO** forces acting on the metal block P. (2)
- Calculate the mass of the metal block P, if the weight recorded by the spring balance is 2 N. (2)
- Load P was removed, and another load Q was attached to the spring balance as shown below. It was noticed that the spring balance now extended to 3N.

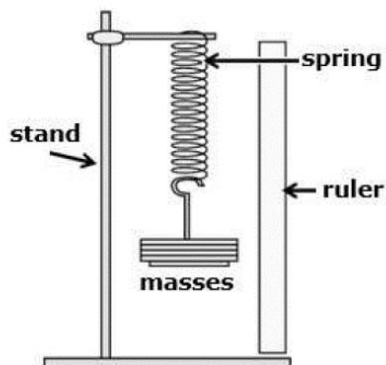


- Calculate the extra extension experienced by the spring when Q is attached instead of P. (1)
- What will the reading of the pointer be if no load is attached to the spring balance? (2)
- Calculate the new reading of the pointer if both loads Q and P are hung together to the spring. (2)
- State the law used to calculate your answers above. (1)

2. May 2019 Paper I

(10 marks)

Isaac and Anna have been asked by their teacher to carry out an experiment to verify Hooke's Law.



- a. They set up the apparatus as shown in the diagram above.
 - i. State Hooke's Law
(2)
 - ii. When they start the experiment, Isaac finds it difficult to take measurements accurately from the ruler with the apparatus given and suggests adding an extra piece of apparatus. What other apparatus do they need? (1)
 - iii. Copy the diagram above and include this extra apparatus in the diagram. (1)
- b. State **ONE** precaution they should take to ensure accurate results. (1)
- c. They noted the initial length of the spring without any mass loaded to it. Each time a new mass was added, a new length of spring was recorded. Complete the two missing headers in the spaces (i) and (ii) below, to show the **TWO** other quantities required for this investigation. Include **BOTH** quantities and units. (2)

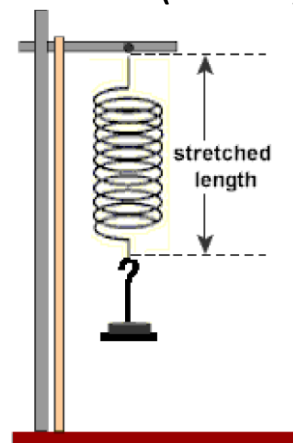
Mass/kg	(i)	New length of spring/m	(ii)

- d. They noted that the length of the spring without any mass loaded to it was 5.0 cm. With a mass of 0.2 kg loaded to it, the spring's new length is 12.5 cm. Assuming Hooke's Law is obeyed, calculate the new length of the spring when a mass of 0.5 kg is used. (3)

3. September 2018 Paper I**(10 marks)**

Theresa was carrying out an experiment investigating the stiffness of a spring.

- Give the name of the law that she needs to use. (1)
- What does the law state? (2)
- Theresa suspended a 10 cm long spring vertically downwards and attached a 100g mass to it. Calculate the force applied to the spring. (2)
- When Theresa measured the length of the spring, she found that it was 16 cm long. What is the value of the spring constant? (3)
- What would the new length of the spring be if a 150 g mass was suspended from the spring? (2)

**4. September 2017 Paper I****(10 marks)**

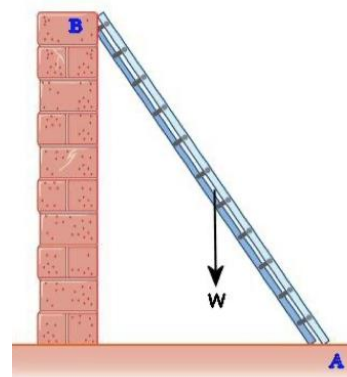
A force is defined as a pull or a push that one object exerts on another. There are various types of forces. The ladder, shown in the diagram below is resting against a wall. This question is about the several forces acting on the ladder.

- Underline the correct word (scalar or vector) and fill in the **TWO** blank spaces in the following sentence.

A force is a (scalar/vector) quantity because it has both _____ and _____ (3)

- Lisa wants to hang a picture up a wall and uses a ladder AB of mass 12kg to do so. The ladder is standing on rough ground and resting against the wall. There are several forces acting on the ladder. One of them is the pull of the earth on the ladder, commonly known as weight. This force is labelled W in the diagram.

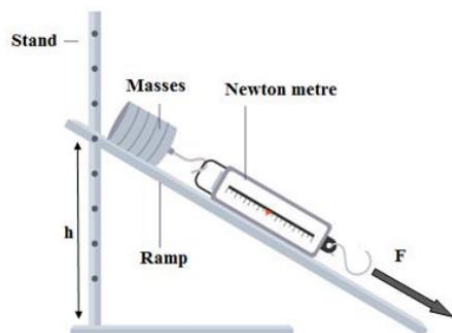
- Calculate the size of this force W. (2)
- Draw on the diagram an arrow, labelled R, showing the vertical reaction force of the ground on the ladder at A. (1)
- Lisa now starts to climb up the ladder. How would this affect the size of force R? (1)



- The base of the ladder does not slip and move away from the wall because of the frictional force between the ladder and the ground at A.
 - On the diagram draw an arrow, labelled F, showing this force on the ladder. (1)
 - Circle the correct answer:
If the ground was smooth and not rough, the size of this frictional force would have been smaller/the same/greater. (1)
 - For the same ladder and type of ground, name **ONE** way of reducing the chance of the ladder slipping away from the wall. (1)

5. April 2016 Paper I**(10 marks)**

Alan wanted to investigate the frictional forces acting on a mass being pulled down a ramp, inclined at different gradients.



(Diagram adapted from <http://www.bbc.co.uk/education/guides/zttfyrd/revision/5>)

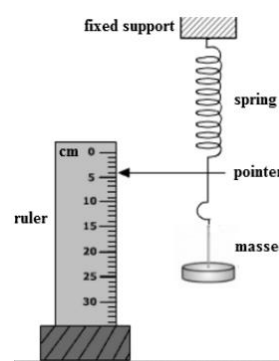
He used a stand to hold the ramp inclined at a vertical height, h , as shown in the diagram above. He then placed the masses at rest at the top of the ramp and attached a Newton metre to the masses. He recorded the force F needed to start moving the masses at different values of h . The following results were obtained.

Vertical height / h (m)	0.05	0.15	0.20	0.25	0.30	0.35
Force / F (N)	23.0	18.5	16.0	13.5	10.5	8.5

- Draw an arrow on the diagram above to show the direction of the force of friction acting on the masses. Label this arrow as 'friction'. (1)
- Plot a graph of force F on the y-axis against height h on the x-axis. (4)
- What can you conclude about the relationship between the vertical height of the ramp and the force required to start the mass moving? (1)
- How is the force of friction changing as the vertical height is increased? Explain your answer. (2)
- What force is required to start the mass moving when the ramp is at a vertical height of 0.10 m? (1)
- What can Alan do, other than changing the value of h , to reduce the frictional force acting in this situation? (1)

6. April 2016 Paper I**(10 marks)**

- State Hooke's law. (2)
- The spring shown below is loaded with a mass of 1.00 kg, and the pointer moves from the 0 mark to the position shown
 - Read the extension of the spring from the ruler and give your answer in metres. (1)
 - Provided the elastic limit is not reached, where would the position of the pointer on the ruler be if the mass is tripled? (1)
 - Calculate the force required to obtain an extension of 15 cm. (2)



- Sketch the shape of the graph obtained when loading the spring beyond the elastic limit. On your graph show clearly the elastic limit with an X. (3)
- Explain what will happen to the spring as the elastic limit is exceeded. (1)

Density

7. May 2019 Paper I

(10 marks)

- a. A mineral water plastic bottle contains $2.0 \times 10^{-3} \text{ m}^3$ of water and has a total mass of 2.04 kg.
- Calculate the mass, in kg, of the water in the bottle given that the density of Water is 1000 kg/m^3 . (2)
 - Hence calculate the mass, in kg, of the empty plastic bottle. (1)
 - If around 50 million of these water bottles were to be collected for recycling every year, calculate the volume, in m^3 , of plastic collected yearly. Take the density of plastic used to be 1380 kg/m^3 . (2)
 - Empty plastic bottle collected in bulk for recycling are usually compressed using a hydraulic press so that it takes up less space. What effect does this have on the density of the waste collected? (1)
- b. Choose ONE correct answer for the given situation:
 Lisa puts some ice cubes in a glass of water and notices that ice floats. This happens because the spacing between the particles in ice is:
- smaller than that in water;
 - larger than that in water;
 - at the same distance as that in water.
- c. Some water is poured into a glass, and it is left directly exposed to the Sun. After a few hours, the level of the water is considerably reduced. Explain why and how this happens. (3)
-



8. May 2018 Paper I

(10 marks)

A boy throws a piece of wood measuring 3 cm by 4 cm by 6 cm into a swimming pool.

- Define the term density. (2)
 - What is the volume of the wood in m^3 ? (2)
 - If the piece of wood has a mass of 7 g, what is its density in kg/m^3 ? (3)
 - The piece of wood is now broken in two pieces. What effect, if any, will this have on the density of each part of the wood? (1)
 - Would you expect the wood to float or sink when placed on water? Explain your answer. (2)
-

9. May 2017 Paper I**(10 marks)**

The mass of air in a room can be estimated from its volume and the density of air. A room has sides of 4.5 m, 3.5 m and 4 m. the density of air is 1.1 kg/m^3 .

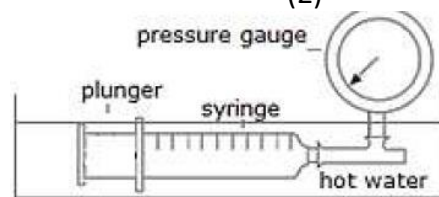
- a. Define the term density. (2)
 - b. What is the volume of the room in m^3 ? (1)
 - c. Calculate the mass of air in the room. (2)
 - d. A heater is placed in the room. Explain, in terms of particles, what happens to the air in the room when the air above the heater heats up. (4)
 - e. Animals exhale carbon dioxide. In a closed room, carbon dioxide collects at the bottom of the room. What does this show about the density of carbon dioxide? (1)
-

Pressure

10. May 2019 Paper I

(10 marks)

- a. A syringe containing air at room temperature is connected to a pressure gauge, which is an instrument that shows the pressure in liquids and gases.
- Explain how the air particles inside the syringe create a pressure on its walls. (1)
 - The plunger is then pulled outward, to increase the volume of the air. State what happens to the reading on the pressure gauge, and explain your answer, in terms of the behaviour of the air particles. (2)
 - The plunger is not locked in position, to keep a constant volume. The syringe is placed in a container of hot water. State what happens to the pressure of air inside the syringe, and explain your answer, in terms of the behaviour of the air particles. (2)
- b. The Earth is surrounded by a layer of air. The table below shows how the value of atmospheric pressure varies with height above the Earth's surface.



Height/km	0	5	10	15	20	25
Pressure/ kPa	101	54	27	12	5.5	2.5

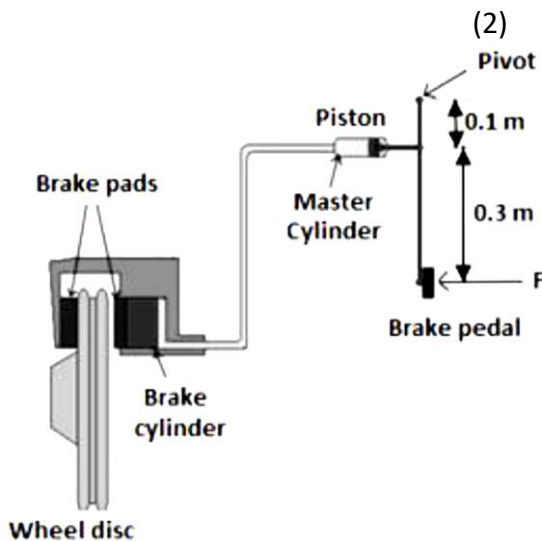
- State the value of atmospheric pressure, in Pa, at a height of 5 km. (1)
- State what happens to the atmospheric pressure with increase in height above the Earth's surface. Give **ONE** reason why this happens. (2)
- Aircraft normally fly at a height of about 10 km. The inside of the aircraft is kept at atmospheric pressure. Explain why the doors and windows of the aircraft need to be airtight. (2)

11. September 2019 Paper IIB

(20 marks)

- a. A footballer of mass 70 kg is wearing a pair of ordinary shoes. The total area of both shoes in contact with the ground is 0.04 m^2 .
- Define pressure. (1)
 - Calculate the pressure he exerts on the ground when standing on both feet. (2)
 - This footballer then puts on a pair of football shoes, shown in the diagram on the right. Explain in terms of pressure why the several studs at the bottom of these shoes help to increase grip with the playing surface. (2)
- b. A sea diver is diving at a depth of 30 m in the sea.
- Calculate the pressure that the sea water is exerting on him given that the density of sea water is 1025 kg/m^3 . (2)
 - If atmospheric pressure on this day is 101 kPa, calculate the total pressure on the diver. (1)



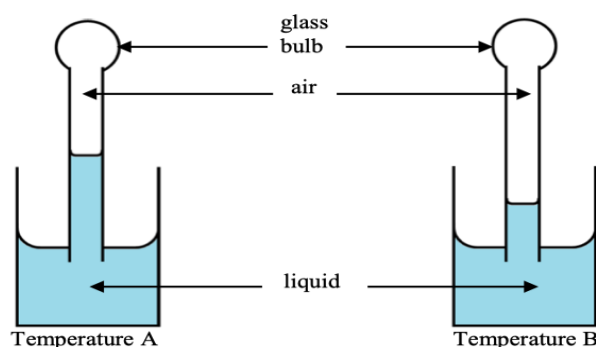
- iii. The air bubbles released by the diver at this depth are seen to increase in volume considerably while they are rising to the surface. Explain why this increase in volume occurs. (2)
- c. The diagram below shows a simplified view of the hydraulic brake system of a car. Only one wheel and one brake cylinder is shown for clarity. The brake pedal is pivoted at the top and acts as a lever. When a force is applied to the pedal as shown, the piston in the master cylinder is pressed inwards. The pressure exerted on the oil in the system forces the piston in the brake cylinder to move inwards. This action causes the brake pad to press against the wheel disc which in turn slows down the car.
- 
- i. A force F of 20 N is applied to the brake pedal. Use moments to calculate the force that this produces on the piston in the master cylinder. (2)
- ii. The area of the piston in the master cylinder is $4 \times 10^{-5} \text{ m}^2$. Calculate the pressure exerted on the oil by the piston in the master cylinder. (2)
- iii. What pressure is exerted by the oil on the piston in the brake cylinder? (1)
- iv. The area of the piston in the brake cylinder is $1.0 \times 10^{-4} \text{ m}^2$. Calculate the force that this piston exerts on this one brake pad. (2)
- v. Give ONE advantage of using oil instead of water as a liquid here to transmit pressure. (1)
- vi. If some air manages to find its way inside the oil in this system, what effect would this have on the braking force as calculated in part (iv)? Explain. (2)

12. May 2017 Paper IIA

(20 marks)

This question is about pressure.

- a. The thermoscope was a device built in approximately the year 1593 to indicate temperature. Two diagrams of a thermoscope at different temperatures are shown below.



By referring to the theory of particles, explain which of the temperatures, Temperature A or Temperature B, if any, is higher. (3)

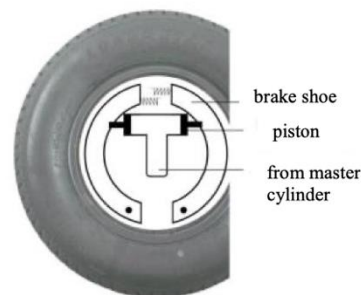
- b. In a car drum-brake, a master cylinder of area 5cm^2 leads to two pistons each with an area of 0.0031m^2 .

i. If a force of 10N is applied on the master cylinder, calculate the pressure in the hydraulic fluid in Pa. (2)

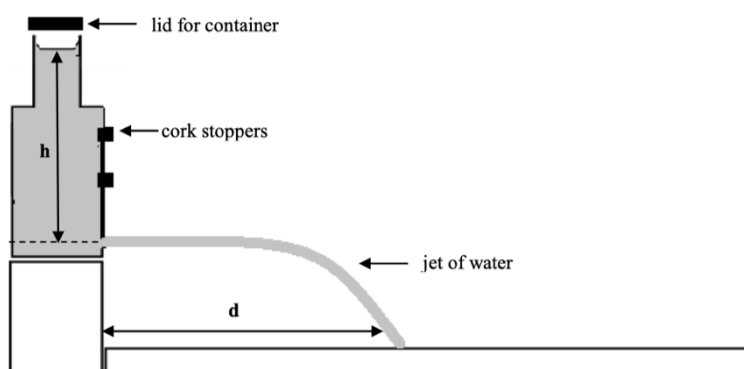
ii. What is the pressure in the hydraulic fluid leading to one of the two pistons? (1)

iii. Calculate the total force exerted by the two pistons. (2)

iv. If the master cylinder is connected to two pistons for each of the car's four wheels (a total of eight pistons), how does this affect the value of the force calculated in part b(iii)? (2)



- c. A group of students investigate how the pressure of water varies with height by measuring the maximum distance reached by the water, d , on changing the height occupied by the water, h , in the apparatus shown below.

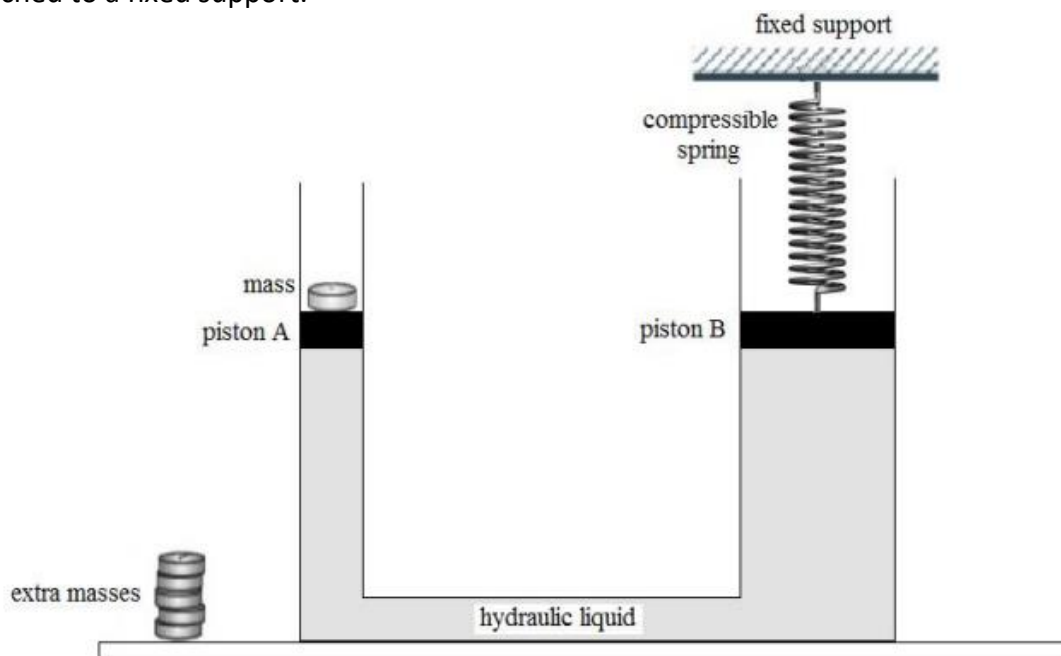


- i. Predict, giving reason/s, what would happen to the distance, d , reached by the water if:
- The height, h , is increased;
 - Ethanol, with a density of 0.8g/cm^3 , is used instead of water;
 - The container is sealed with a lid.
- (6)
- ii. If water has a density of 1000kg/m^3 and the hole has an area of $2 \times 10^{-5}\text{m}^2$, calculate the force by which the water shoots out of the hole when the height, h , is 0.15m . (4)

13. September 2016 Paper IIB

(20 marks)

- a. The figure below shows a simple hydraulic machine. A number of masses can be placed on piston A. Piston B is attached to a compressible spring, which in turn is attached to a fixed support.



- i. Draw an arrow on the figure above to show the force acting on piston A when a mass is placed on it. (1)
 - ii. Draw an arrow on the figure above to show the force acting on piston B when the mass is placed on piston A. (1)
 - iii. The student measures the diameter of piston A and piston B. Explain how the student can find the area of the two separate pistons. (2)
- b. The student reads some information about hydraulic machines and finds the statement below.
- i. Fill in the blanks. (2)
 The _____ on piston A is _____ to the pressure on piston B, as it is _____ equally through the _____.
 - ii. The student decides to investigate the statement in part (i) by placing different masses on piston A and then measuring the compression in the spring attached to piston B. He realises that a metre ruler, a pointer, a stand and a clamp are required. Include this apparatus with the diagram and label it clearly. (2)
 - iii. Put the following statements in order by using the numbers 1 to 7, to describe the method that the student should follow to investigate the statement in part(i). The first one is done for you. (3)

He places a mass on piston A and finds the force acting by using $W = mg$.	
He records the new position of piston B using the pointer on the ruler and calculates the compression.	
The diameter of both pistons is measured, and their cross-sectional area is calculated.	1
He repeats this procedure using several masses to investigate if the statement is true for all forces applied at A	
Given that the spring compresses by 0.5 mm for every 1.0 N of applied force, he can calculate the force on piston B.	
He records the original position of piston B using the pointer and ruler.	
The pressure on both pistons can be calculated by using $P = F/A$.	

- c. Given that a mass of 100 g is placed on piston A, calculate the force acting on piston A. (2)
- d. The area of piston A is 0.00126 m^2 . Calculate the pressure on piston A. (2)
- e. State the value of the pressure on piston B. (1)
- f. Given that the area of piston B is 0.01540 m^2 , calculate the force acting at piston B. (2)
- g. If the spring used can compress by 0.5 mm for every 1.0 N, calculate the compression of the spring in mm. (2)
-

14. April 2016 Paper I**(10 marks)**

John had a small column in his house entrance and wanted to buy a statue to place on it.

- a. He went to his local shop and found that the statue he liked most was available in two different materials. This is the information that he was given from the shop attendant.

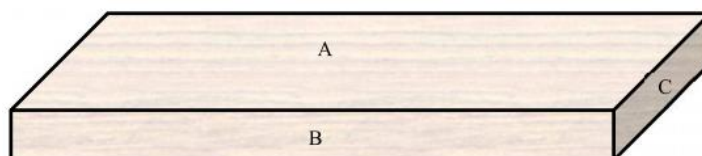
	Statue A	Statue B
Material	Bronze	Marble
Density	8900 kg/m^3	2560 kg/m^3
Volume	0.027 m^3	0.027 m^3

- i) What does the value of density represent? (2)
- ii) Calculate the weight of statue B. (2)
- b. John did not know if the column could support such weights, so he went back home to do some research. He found out that the maximum pressure that it can withstand without breaking is 27500 Pa .
- i. The base of the statue is circular and has a radius of 0.15 m. Calculate the area of contact of the statue with the column. (1)
- ii. Calculate the maximum weight of such a statue that the column can hold without breaking. (2)

- iii. Based on your answer to b(ii), which statue should John buy in order not to damage the column? (1)
- c. John bought another small statue to place in his fish tank as a decoration. Calculate the pressure acting on the statue due to a water column of 1.20 m. The density of water in the fish tank is 1022 kg/m^3 . (2)

15. April 2015 Paper I**(10 marks)**

- a. Complete the following:
- Mass is the (1)
 - Weight is the (1)
- b. A long wooden plank has a mass of 450 g.



- Calculate the weight of the plank. (2)
- Copy the diagram and mark with a letter W the point where all the weight seems to act. (1)
- What is this point called? (1)
- When this plank rests on the ground, its area of contact with the ground is 1.4 m^2 . Calculate the pressure that the beam exerts on the ground. (2)
- On which face does one have to place the wooden plank in order to exert maximum pressure? Explain. (2)

16. April 2015 Paper I**(10 marks)**

Robert Boyle was an Anglo-Irish physicist and inventor. During the 17th century he studied the behaviour of gases.

In one of his experiments he used a fixed mass of gas and started to apply pressure to decrease its volume. He observed what happened to the gas pressure and noted the values obtained in the following table.

Pressure, P (kPa)	20	30	40	50	60
Volume, V (cm ³)	250	155.67	125	100	83.33
1/Volume, 1/V (1/cm ³)	0.004				

- Copy and complete the table above. (2)
- Draw a graph of **1/Volume** on the x-axis against **Pressure** on the y-axis. (4)
- What is the relationship between Pressure and 1/Volume? (1)
- Explain what would happen to the Pressure when the Volume is halved. (1)
- Explain in terms of molecular motion the cause for this change in pressure when the volume of a fixed mass of gas is halved. (1)
- What should be assumed in part (v)? (1)

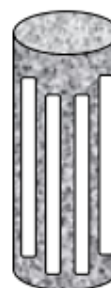
17. September 2015 Paper I**(10 marks)**

Quantities in science are classified as scalar or vector.

- State the difference between a vector and a scalar quantity. (2)
- Which of the following quantities are vectors or scalars? (The first one has been done for you.) (3)

Quantity	Vector	Scalar
Distance		X
Velocity		
Kinetic Energy		
Weight		
Displacement		
Speed		
Momentum		

- Engineers were planning to install decorative columns made of marble, of density 2560 kg/m^3 , in front of a building. The columns were 2.72 m tall and of radius 45 cm.
 - Calculate the mass of one column. (2)
 - Calculate the pressure exerted by each column on the ground. (2)
 - Engineers found out that the ground beneath the columns would not withstand the pressure calculated in part c(ii). Suggest one possible change to each column, the engineers can do, to solve this problem. (1)

**18. September 2015 Paper I****(10 marks)**

When installing glass panes, glass fitters use rubber suction cups (suckers).

- Define atmospheric pressure. (2)
- Explain how atmospheric pressure acts when glass fitters are using the rubber suction cups. (2)
- With the aid of a diagram and with reference to atmospheric pressure, describe how a person can suck a liquid from a container using a straw. (3)
- Another instrument used to measure atmospheric pressure is the mercury barometer. What happens to h when the atmospheric pressure increases? (1)
- What would happen to the mercury level if the barometer was gradually moved to outer space? Why? (2)

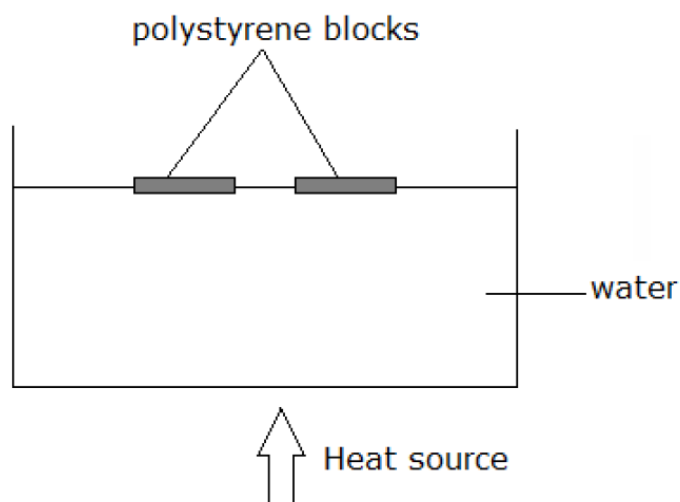


Heat

19. September 2020 Paper I

(10 marks)

A teacher sets up a metal water tank with 15 kg of water at a temperature of 21 °C, to explain plate tectonics, which describe the movement of the Earth's continents. Two blocks of polystyrene float on top. The polystyrene blocks do not move.



The water tank is placed on an electric hot plate and the heat is turned on.

- By which process is the heat transferred through the water. (1)
 - Explain, in terms of the density of water, how this process happens. (2)
 - polystyrene blocks start moving. In which direction do they move? (1)
 - The hotplate has a power rating of 8190 W and is left on for 5 minutes. Calculate the energy supplied to the water. (2)
 - Calculate the final temperature of the water, if water has a specific heat capacity of 4200 J/kg°C. (3)
 - In order to verify their answer, the students measured the final temperature of water. They
 - found it to be lower than that calculated above. Give ONE reason for this difference. (1)
-

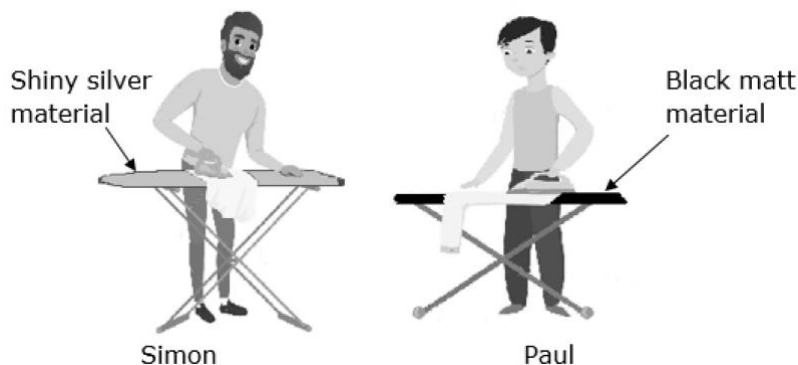
20. September 2020 Paper IIA

(20 marks)

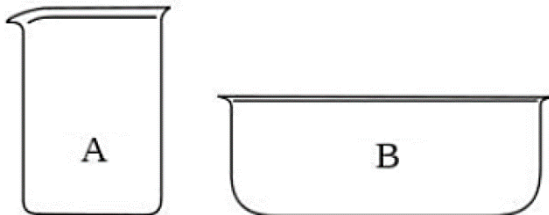
- Paul and Simon hang their cotton school T-shirt to dry after it has been washed. The diagram below shows how the two men hung their T-shirt.



- i. By which process is the heat transferred from the sun to clothes hung on the line. (1)
 - ii. Mention and explain the process by which the T-shirts will dry. (2)
 - iii. Which T-shirt will be the one to dry first? Give a reason for your answer. (2)
 - iv. How does the drying time change if a breeze starts blowing? Explain your answer. (2)
- b. Paul and Simon iron their T-shirt as soon as it dries. They use two different ironing boards. Paul's ironing board is covered in a dark matt material, whereas Simon's is covered in a shiny silver material.



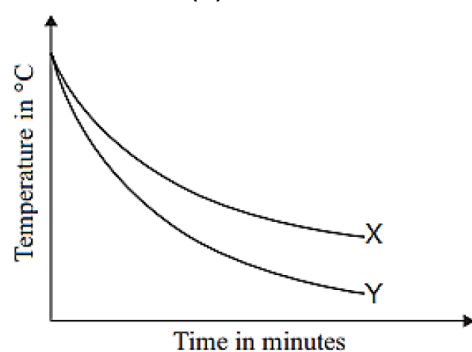
Which iron board is more efficient in transferring the heat onto the T-shirt so as to iron out all the creases? Explain your answer. (2)

- c. Paul and Simon want to investigate the rate at which hot water cools down in two glass containers with two different surface cross-sectional areas. They were provided with containers A and B.
- 
- i. Mention TWO additional pieces of apparatus required to carry out this experiment. (2)
 - ii. Briefly describe how they should carry out the experiment. (4)
 - iii. Mention **TWO** factors that they should keep constant so that they would have a fair and reliable result. (2)
 - iv. Simon and Paul plotted graphs X and Y of temperature in °C against time in minutes for both containers. Identify which graph belongs to container A and which graph belongs to container B. Explain how you arrived at your conclusion. Copy the below:

Graph X = _____

Graph Y = _____

Reason:



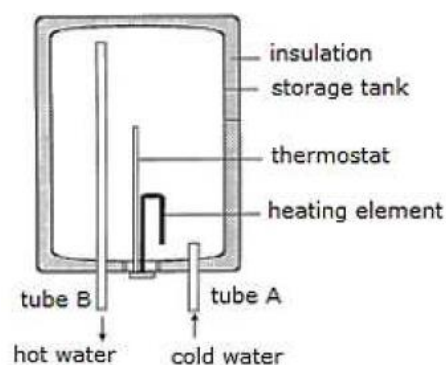
_____ (2)

- d. The fennec fox lives in the hot Sahara Desert. It has very large ears through which it loses heat. Explain why this feature is important for an animal that lives in the desert. (1)



21. May 2019 Paper I**(10 marks)**

The electrical water heater shown in the diagram contains 80 kg of water. Cold water at 20 °C enters from tube A and is heated by the heating element. The hot water produced leaves the heater from tube B.

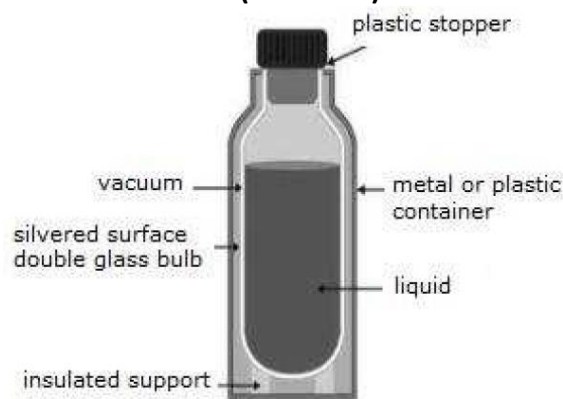


- Show, using an energy flow diagram, the energy changes occurring in the heating element. (1)
- Hot water flows from the bottom to the top inside the tank. What is this process called? (1)
- Calculate the energy required to heat the water in the tank to 60 °C, given that the specific heat capacity of water is 4200 J/kg°C. (2)
- Calculate the time taken to heat the water, given that the power of the heater is 1500 W. (2)
- In practice, the time required is longer than the amount calculated in part d) as not all the heat produced by the heating element is absorbed by the water. Give **TWO** reasons for this, stating clearly where the 'lost' energy is being transferred to. (2)
- A solar water heater could also be used to heat water in our households.
 - State ONE advantage of using this method. (1)
 - Why are some surfaces of a solar water heater painted black? (1)

22. May 2019 Paper IIA**(20 marks)**

This question is about energy transfer.

- Lisa is going out on a picnic and she prepares a hot drink in a vacuum flask, designed to keep liquid warm for some time. The diagram shows the inside of such a flask.



(<https://www.123rf.com>)

State which method of heat transfer is being
Reduced by the parts of the flask listed below:

- Glass bulb (one method); (1)
 - Silvered surface (one method); (1)
 - Plastic stopper (one method); (1)
 - Vacuum between walls (two methods) (2)
- During the Physics lesson Lisa is told that there are also vacuum flasks with an inside bulb made of stainless steel instead of glass. She wants to carry out an experiment to investigate which of the two flasks keeps liquid warm for a longer time. She is provided with a stainless steel flask and a glass flask of the same size, two thermometers, a stop watch and hot water.
 - Describe how she should carry out the experiment (4)
 - State **TWO** important precautions necessary in this experiment. (2)

- iii. The table below contains the three columns used to record the readings taken. Copy the table and fill in the other **TWO** column headings, with the quantities measured and their units. (2)

Time/s		

- iv. Draw a rough sketch (with labelled axes) of the two graphs expected for the two materials. You are **not** expected to identify which graph is for glass and which graph is for stainless steel. (3)
- v. Why is it better to plot the two graphs on the same axes and not on separate ones? (1)
- vi. How can Lisa use her graphs to decide which flask is better at keeping hot liquids warm for a longer time? (1)
- c. Vacuum flasks are also used to keep certain liquids at very low temperatures. In industry, they can be used to keep liquid air at -200°C . Explain, in terms of heat transfer, how a flask can keep cold liquids at low temperatures. (2)

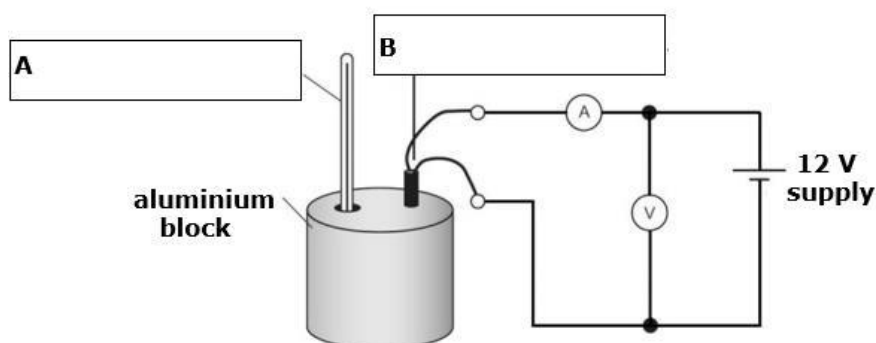
23. September 2019 Paper IIB

(20 marks)

- a. All matter is made up of particles. The following table shows the similarities and differences in the particle properties between solids, liquids and gases. The ones for liquids have been filled in for you. Copy the table and fill in the remaining properties for solids and gases. (6)

	solids	liquids	gases
spacing between particles		closely packed but not in order	
forces between particles		strong	
motion of particles		vibrate but also move randomly throughout the liquid	

- b. A group of students are carrying out an experiment to determine the specific heat capacity of an aluminium block, using an electrical heater. They set up the apparatus as shown.



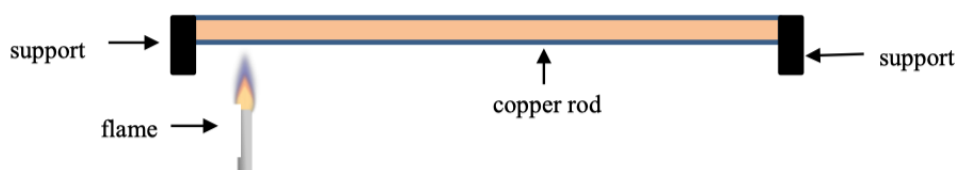
- i. Look at the diagram above and name the two pieces of apparatus A and B. (2)
 - ii. They will need a stopwatch and one other measuring instrument to carry out the experiment. Name this instrument. (1)
 - iii. How is the energy supplied to the block calculated from the stopwatch reading and apparatus A? (1)
 - iv. The teacher advised the students to put lagging around the block. Name a material which can be used for this purpose. (1)
 - v. What is the function of the lagging around the block? (1)
 - vi. Copy the text, inserting the correct word:
Without lagging, the calculated value of specific heat capacity would be smaller than / equal to / greater than the actual value (1)
 - vii. Name ONE other precaution, not already mentioned, which is necessary to obtain an accurate result and explain how it achieves its purpose. (2)
- c. Luca and Sarah are out camping with their families close to the seaside.
- i. Sarah goes swimming and notices that when she comes out of the water, she feels a bit cold until the water still on her body dries up. Explain why cooling occurs as a liquid evaporates. (2)
 - ii. Luca had two buckets. One of them was much wider than the other but they both had the same volume of water in them. He observed on the following day that more water evaporated from the wide bucket than from the narrower one. Explain why this occurred. (1)
 - iii. Sarah had learned at school the fact that if the same amount of heat energy is given by the sun to 1 kg of sand and 1 kg of water, the sand heats up much more than the water. What can she say from this about the values of the specific heat capacity of each? (1)
 - iv. Why is it that after sunset the sea takes a longer time to cool down than the sand? (1)

24. May 2018 Paper IIA**(20 marks)**

This question is about heat energy.

- a. The kinetic theory of matter explains the structure of matter. Fill in the missing words:
 - i. In a solid, the particles are _____ to each other and _____ about a mean position. (2)
 - ii. In a liquid, the particles can _____ around and are further away from each other from the particles in _____. (2)
 - iii. In a gas, the particles are very _____ from each other. (1)

- b. A copper rod fixed between two rigid supports is heated uniformly.



- Describe why the support will experience a force when the copper rod is heated? (2)
 - With reference to particles, explain the effect of the rod in part (i). (2)
 - What can happen to the rod if it cannot move the supports? (1)
- c. The temperature of the copper rod was raised by 50°C while heating. If the mass of the copper rod is 750 g and the specific heat capacity of copper is $385 \text{ J/Kg}^{\circ}\text{C}$, calculate how much heat was given to the rod. (3)
- d. The three main methods of heat transfer are conduction, convection and radiation.

Copy and complete the following table:

Type of Heat Transfer	Can take place in:	Method of how heat is transferred:
Conduction		
	Liquids or gases	
		Electromagnetic waves

- When birds are about to sleep they often flutter their feathers giving the appearance of a ball of feathers. Explain in terms of physics why they do this. (3)
- Give ONE method of how we can insulate our homes. (1)



25. September 2018 Paper I

(10 marks)

A small iron wheel is to be fitted with an iron ring around it.
The mass of the ring is 500 g and the volume is 64 cm^3 .



- One way how this can be done is by first heating the ring. Explain how the ring ends up fitted tightly around the wheel. (2)
- Would the density of the ring change when it is heated? Explain your answer. (2)
- One way to cool down the wheel with the hot ring is to spray it with water which fizzles off. Explain how this brings about cooling. (2)
- What is the density of the ring in g/cm^3 ? (2)
- If the wheel and ring are dropped into a container of water, they sink. Explain. (2)

26. September 2018 Paper IIB**(20 marks)**

This question is about specific heat capacity.

Katrina and Bjorn wanted to determine the specific heat capacity of water and cooking oil. The equipment available was a small plastic container, a mass balance, an electric heater, a thermometer and a joule meter.

- a. Draw a diagram of how this equipment may be set up. (3)
 - b. Katrina and Bjorn used the equipment to carry out the experiment and presented the following report of the method used, but the steps are in the wrong order. Rewrite the method in the correct order. (6)
 - The final temperature and the reading of the joulemeter were taken.
 - The heater was switched off.
 - The mass of the liquid and container was measured using the balance.
 - The initial temperature was measured using the thermometer.
 - The electric heater was switched on.
 - The mass of the container was measured using the balance.
 - c. Mention ONE important precaution that needs to be taken. (2)
 - d. If during the experiment, the temperature of 50 g of water was raised by 2°C , what reading did the joulemeter give if the specific heat capacity of water is $4200 \text{ J/Kg }^{\circ}\text{C}$ (2)
 - e. On a different day, Katrina and Bjorn went for a day at the beach. During the day, they noticed a cool breeze coming from the sea. Katrina gave the following explanation to Bjorn about the formation of this breeze. Complete the missing words. (4)

During the day, the land warms up _____(i)_____ than the sea because it has a smaller _____(ii)_____. As a result, hot air _____(iii)_____ over land which has to be replaced by air coming from the _____(iv)_____. This moving air is the sea breeze that Katrina and Bjorn feel.
 - f. In the evening, after sunset, Katrina and Bjorn noticed that the breeze had changed direction. Fill in the missing words. (4)

In the evening, the land _____(i)_____ heat faster than the sea. The air over the sea is therefore _____(ii)_____ and moves _____(iii)_____. The air has to be replaced by air from land which explains the change in direction of the _____(iv)_____.
-

27. May 2017 Paper I**(10marks)**

In an experiment to estimate the specific heat capacity of brass, a thermometer and a 450W heater were placed in a 1.4kg block of brass with an initial temperature of 20°C. the table below shows part of the results obtained.

Temperature, $T(^{\circ}\text{C})$	25	30	35	40	45
Change in Temperature, $\Delta T(^{\circ}\text{C})$	5	10	15		25
Time, $t(\text{s})$	6	12	18	24	30
Total Heat Energy Supplied, $H(\text{J})$	2700	5400	8100		13500

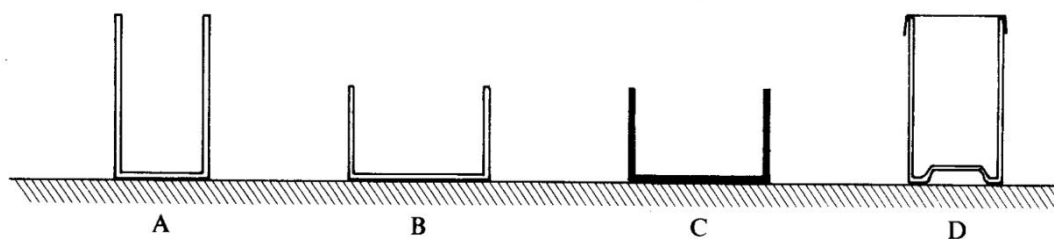
- Fill in the **two** missing values in the above table. Show your working. (3)
- Plot a graph of Total Heat Energy Supplied, $H(\text{J})$ on the y-axis against Change in Temperature, $\Delta T(^{\circ}\text{C})$ on the x-axis. (4)
- Calculate the gradient of the graph. (2)
- Give **ONE** precaution taken in this experiment. (1)

28. September 2017 Paper IIB**(20 marks)**

This question is about heat and heat transfer.

It is common knowledge that a cup of hot drink left on a table cools down and so warms its surroundings.

Sarah works for a company which manufactures plastic cups used to hold a drink of hot coffee. The company has four different cup designs as shown in the following figure.

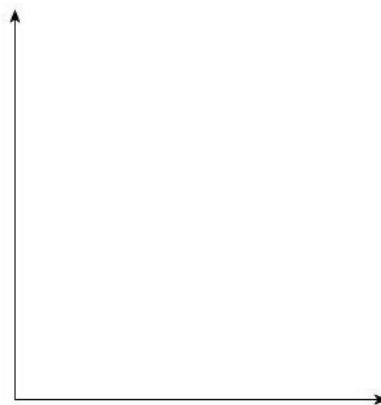


Cup C is made from dark plastic while cups A, B and D are made from white plastic. The base of cup D is elevated at the center, leaving an air space between the cup and the surface; this cup is also fitted with a lid. All 4 cups have the same volume. The manufacturer wants to know which cup can keep the coffee hot for the longest time. He asks Sarah to design an experiment to find this out. Sarah fills the cups with hot water and measures the temperature of the water in each cup at one minute intervals.

- a. To prepare the hot water needed for this experiment, Sarah fills an electric kettle with 1.5kg of cold water at a temperature of 15°C and switches on the kettle until the water boils at 100°C. The specific heat capacity of water is 4200J/kg°C.
- Calculate the energy required to bring the water to the boil. (3)
 - Calculate the time it takes for the water to boil if the power of the kettle is 2000W. (2)
 - Explain why in practice the time taken is longer. (2)
- b. Having obtained the hot water, Sarah then carries out the experiment. The steps needed in the method she used are shown below but are not in order. In the space on the right-hand side, write down the numbers 1 to 5 to show the correct order. 1 being the first step whilst 5 is the last step. (5)

The results are recorded in a table.	
Each cup is filled with the same volume of hot water	
A graph of temperature against time for each cup is plotted	
The four empty cups are placed on the same surface	
The temperature of the water in each cup is taken at regular intervals	

- c. State **ONE** precaution that needs to be taken before taking each temperature and briefly explain why this is necessary. (2)
- d. Draw a rough sketch of the graph you would expect to obtain for any one of the cups. It is important to label both axes with the quantities. (2)
- e. Give a reason why:
- The water in cup B is found to remain hotter than in cup C. (1)
 - The water in cup A is found to remain hotter than in cup B. (1)
 - Give **TWO** reasons why the water in cup D is expected to remain hotter than in cup A. (2)

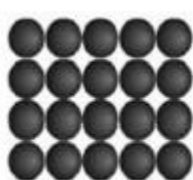


29. September 2016 Paper I

(10 marks)

The figure shows a stainless steel saucepan on a gas burner. Food is being cooked in boiling water in the saucepan

- Name the process by which heat is transferred through the **saucepan**. (1)
- Name the process by which the **water** is being heated. (1)
- The figures below show three different ways in which particles in a substance may be arranged.



A



B



C

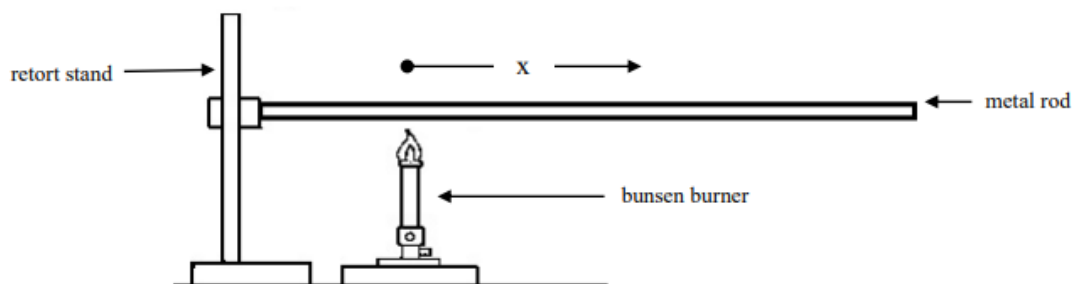
Indicate which figure A, B or C, best represents the arrangement of the particle

- In the **stainless steel saucepan**. (1)
 - In the **steam** coming out of the saucepan. (1)
- Give one reason why the saucepan is made of stainless steel. (1)
 - Suggest a suitable material for the handle of the saucepan. Explain. (2)
 - Evaporation can be described as the _____ from the _____ (2)
 - A lid is now used to cover the saucepan. Give one reason why this lid is needed. (1)

30. September 2016 Paper I

(10 marks)

The temperature along a metal rod was measured at equal intervals from one end which was being heated using a Bunsen flame.



The temperatures measured were recorded in the table below

Distance (x)/cm	0.0	5.0	10.0	15.0	20.0	25.0
Temperature/°C	160.0	120.0	80.0	40.0	20.0	20.0

- Plot a graph of temperature (y-axis) against distance (x-axis). (4)
- Determine the gradient of the graph till the temperature reached 20.0°C. (2)
- What does the shape of the graph show? (2)
- From the graph estimate a possible value for room temperature. Explain how you arrived at your conclusion. (2)

31. September 2016 Paper I**(10 marks)**

During a typical Maltese summer, concrete roofs and walls are exposed to the sun for a large number of hours. They will absorb heat and the temperature inside the house can increase to uncomfortable levels. On the contrary in winter, the temperature inside the house can reach very low levels.

- Suggest two main precautions that can be taken while building houses in order to reduce the absorption of heat in summer and the heat losses in winter from walls and roofs. (2)
- Double-glazed windows reduce heat losses from the house in winter. Explain. (2)
- Mention any other method in which we can insulate our houses. (1)
- Anne bought a gas heater and placed it in a corner of the living room as shown in the figure below. Add labelled arrows to show how convection currents are heating the room. (2)



- Anne wanted to calculate the amount of energy required to increase the temperature of the room from 8 to 26. She found out that the specific heat capacity of air is approximately 1005.
 - What are the units of specific heat capacity? (1)
 - Calculate the energy required if the mass of air in the room is 278.6kg. (2)

32. April 2015 Paper I**(10 marks)**

Hot air balloons rise up once the air inside them is heated. A balloon which has a volume of 2800 m^3 may be used to lift up two or three people.

- What happens to the volume of the air when it is heated? (2)
- Explain why the balloon rises. (2)
- Explain one possible way of returning the balloon back to the ground. (2)
- Another way of lifting the balloon requires filling it with helium. If the density of helium is 0.164 kg/m^3 , calculate the mass of helium that needs to be used. (2)
- If the balloon goes down at constant velocity, explain in terms of Newton's laws. the forces acting on the balloon. (2)

33. September 2015 Paper I**(10 marks)**

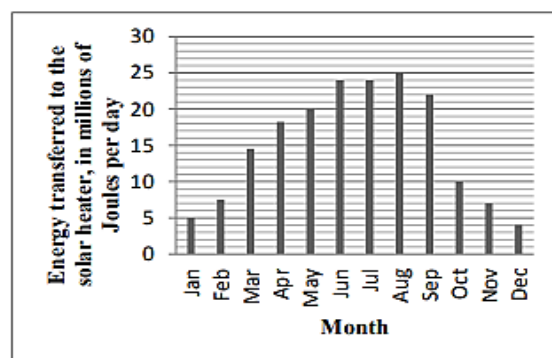
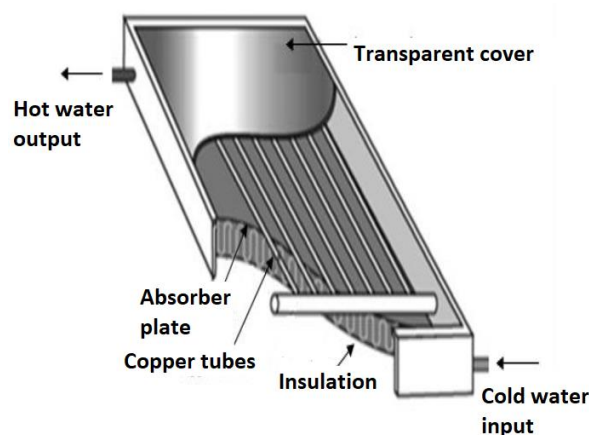
A metal spherical air-filled container is to be heated using solar energy.

- By what process is energy transferred from the sun to earth? (1)
- What colour would you recommend painting the sphere so that it heats up quickly? Explain your choice of colour. (2)
- What happens to the air inside the sphere and to its pressure when it receives energy from the sun? (3)
- Describe in terms of molecules the structure of a solid and that of a gas. (4)

34. April 2016 Paper IIA**(20 marks)**

This question is about heat energy.

- The picture shows one type of solar water heater. Cold water from a tank is slowly pumped through copper tubes inside the solar flat plate collector, which consists of a flat plate absorber, a transparent cover, and an insulated backing
 - How is the energy from the Sun transferred to the flat plate collector? (1)
 - Suggest a colour for the absorber plate located underneath the copper tubes? Explain your answer. (2)
 - State how heat is transferred from the copper tube to the water. Explain why copper is chosen for manufacturing the tubes through which the water flows. (2)
 - Explain why the transparent cover increases the efficiency of the solar heater. (2)
 - What is the purpose of the insulation at the bottom of the flat plate collector? (1)
- In Malta, on average a person uses 30 kg of hot water per day. In order to kill bacteria, the water going into the tank at 20 must be heated to 60 °C. The specific heat capacity of water is 4200 J/kg°C.
 - Calculate the energy needed to increase the temperature of the water consumed to 60°C. (2)
 - Calculate the energy required per day for a household consisting of four people. (1)
 - The bar graph shows the average energy transferred to the solar heater, in millions of Joules, per day for every month of the year. State in how many months of the year all the energy required by the solar heater of a household of four people, can be entirely provided by the sun. (1)

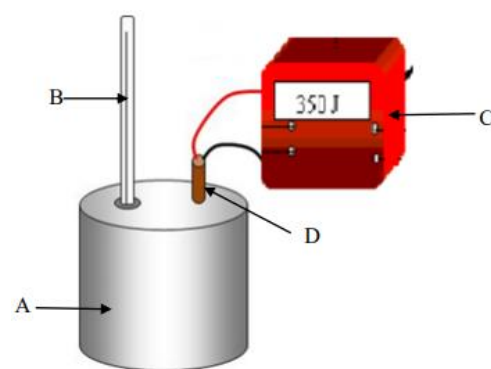


- c. During those months when the solar energy is not enough to heat the water in the tank, an electric heater at the bottom of the tank has to be switched on, until the temperature rises to the necessary value.
- State the name of the process by which the water is being heated inside the tank. (1)
 - Explain clearly how this process is taking place in terms of the water molecules. (3)
 - Should the inside walls of the tank be made of a matt black or a shiny silver coloured material? Explain your answer. (3)
 - State the name of any other object used in our households which should have the same interior, for the same purpose stated in (c iii). (1)

35. September 2015 Paper IIB**(20 marks)**

This question is about specific heat capacity

- a. Trudy wanted to determine the specific heat capacity of an aluminium block. She set up the apparatus shown
- Label all parts – A,B,C and D. (4)
 - The equation which needs to be used is $Q = mc\Delta\theta$. State what each symbol represents. (4)
 - Put the following sentences in the correct order to describe how the experiment is to be carried out. (4)



Record the initial temperature of the metal block.	
Switch off the power supply after a time interval t minutes.	
Plug in the joulemeter and switch it on.	
Calculate the rise in temperature.	
Allow time for the heat energy to spread throughout the metal block.	
Set up the apparatus as shown in the diagram.	
Record the highest temperature of the metal block.	
Record the final joulemeter reading.	
Find the mass of the metal block.	

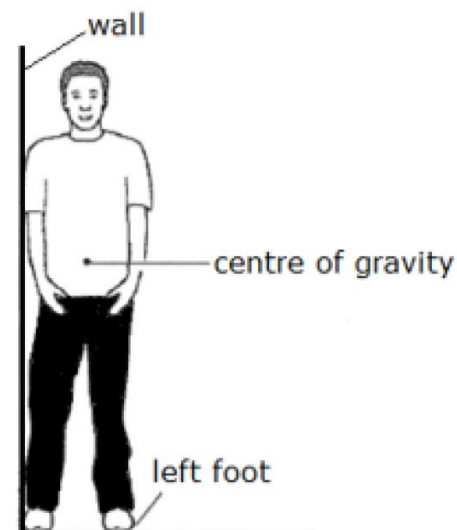
- Mention two precautions that need to be taken during the experiment. (2)
- b. The specific heat capacity of water is $4200 \text{ J/kg}^\circ\text{C}$.
- How much energy needs to be given to 500 g of water so that its temperature rises by 10°C ? (2)
 - If 1 kg of water is used instead, how much energy is required? (2)
 - A teaspoon is placed in the water. After some time, the end of the teaspoon feels warm when touched. Explain. (2)

Moments

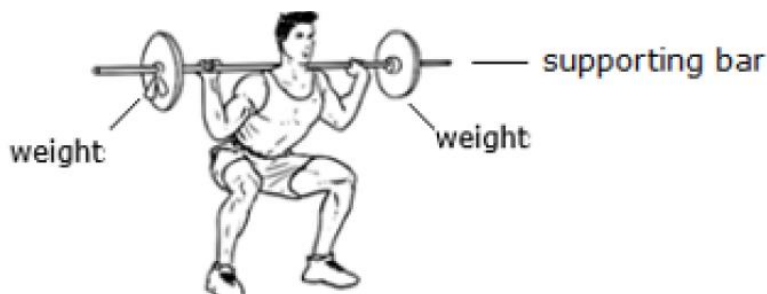
36. September 2020 Paper I

(10 marks)

- a. The diagram shows a boy with his right foot and right shoulder touching a wall. The centre of gravity of the boy lies in the middle of the body as shown.
- Explain 'centre of gravity'. (1)
 - State what happens to the boy if he lifts up his left foot but leaves his right shoulder touching the wall. (2)
 - Give a reason to your answer in part (ii). (1)
 - Will the same reaction occur if the boy had to lift up his right foot instead? (1)



- b. Weightlifting is a sport where force and stability are important. The diagram shows a sportsman of mass 80 kg lifting up weights.

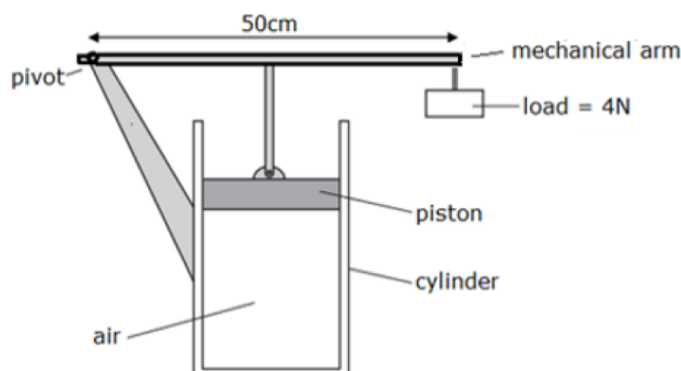


- With reference to the diagram explain **TWO** features that help increase the weightlifter's stability as he lifts up the weights. (2)
- Give a possible reason why it is easier to carry the weight if the centre of gravity of the weights is just behind his neck. (1)
- Calculate the total weight acting on the floor if each weight has a mass of 100 kg and the supporting bar has a mass of 20 kg. (2)

37. September 2020 Paper I

(10 marks)

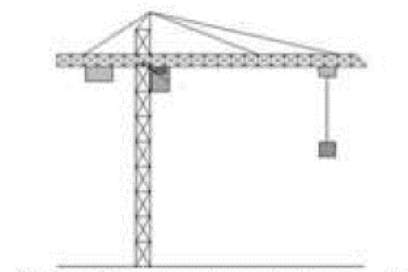
The diagram shows a 50 cm uniform mechanical arm, that has a load of 4 N attached to its end. The piston, located at the centre of the mechanical arm, keeps air particles trapped inside a cylinder.



- a. Describe the movement of the air particles inside the container. (1)
 - b. The system is in equilibrium. Calculate the force exerted by the air particles on the piston. (2)
 - c. The piston has a cross-sectional area of 0.01 m^2 . Calculate the pressure exerted by the air particles on the piston. (2)
 - d. The air inside the cylinder is heated but the system remains in equilibrium.
 - i. State the changes, if any, in the following quantities: (3)
 - Pressure of air inside cylinder
 - Volume of air in the cylinder
 - Density of air in cylinder
 - ii. More heat is supplied until the piston moves upwards. Mention TWO ways how equilibrium can be restored in the system above. (2)
-

38. May 2019 Paper I**(10 marks)**

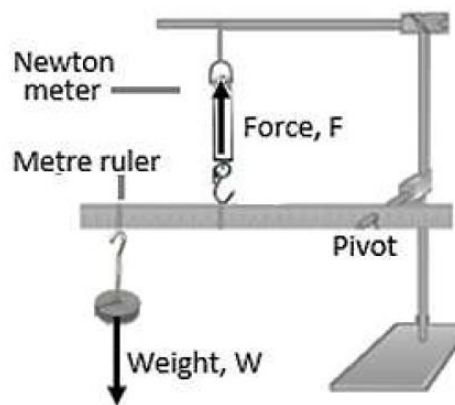
- a. Tower cranes like the one shown in the diagram are common at construction sites. These are used to lift and carry heavy loads, using the principle of moments to keep in equilibrium.



State this principle.

(2)

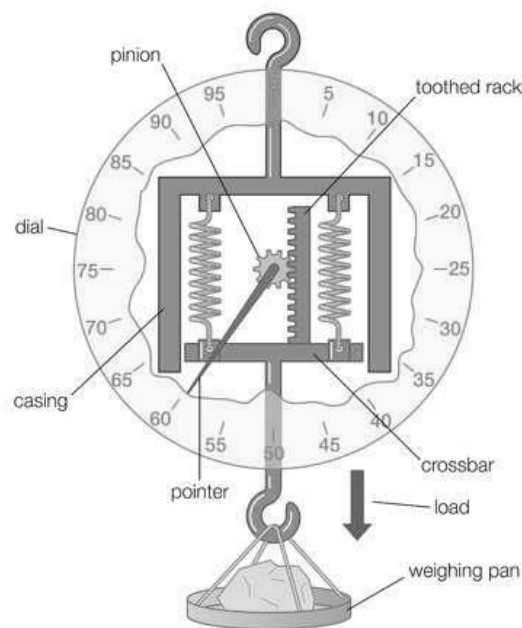
- b. The adjacent set up is used to investigate the principle of moments. A uniform metre ruler is hung from a pivot and is supported by a Newton meter that produces an upward force, F . When a weight W , is hung in the position shown, the ruler is perfectly horizontal.



- i. Include an arrow in the setup to show the ruler's weight. Draw a ruler and label the weight as R . (1)
 - ii. The mass of the ruler used is 0.12 kg . Calculate the moment produced by the weight of the ruler, if it acts at 0.35 m from the pivot. Give your answer in Nm , (2)
 - iii. The value of W is 2.0 N and its distance from the pivot is 0.75 m . Use the principle of moments to calculate the upward force F , produced by the Newton meter to keep the ruler horizontal. The distance between the Newton meter and the pivot is 0.50 m . (4)
- c. If the weight W is moved closer to the pivot, would the Newton meter have to produce a larger or smaller force F , to keep the ruler horizontal? (1)
-

39. September 2019 Paper I**(10 marks)**

The diagram shows a weighing scales operating on the principle of the spring balance. The two springs hang from a fixed casing and their lower ends are attached to a crossbar. As weights are added to the pan, the springs extend, and the toothed rack moves down. This causes the pinion (a round wheel with toothed gears) to rotate and the pointer moves along the dial or scale. The scale is calibrated in Newtons and has a full-scale deflection of 100 N.



- In which direction does the pointer move when weights are added to the pan? (1)
- What is the maximum mass that can be placed on the pan to get full scale deflection? (2)
- Calculate the force in each spring at full scale deflection. (1)
- This instrument is taken to different places on earth. It gives different readings when loaded with the same mass. It would also give different readings on other planets which have a different pull of gravity. Copy the following table and mark with a ✓ whether the reading would be 0, less than or greater than the reading obtained on the surface of the earth. The pull of gravity of the two named planets is stated in brackets for each planet. (3)

	0	less than	greater than
on top of a high mountain on earth			
on the surface of Jupiter (25 N/kg)			
on the surface of Mercury (3.7 N/kg)			

- If the instrument had to have four similar springs instead of two, what load would give full scale deflection of the instrument? Explain. (2)
- What is the advantage of having the pointer attached to this toothed gear system as shown? (1)

40. May 2018 Paper I**(10 marks)**

The concept of moments features in a number of everyday uses.

- a. Define the moment of a force. (2)
- b. A mechanic uses a spanner to undo a tight nut as shown. He places his hand 9 cm from the pivot and applies a force of 10 N.



- i. On the diagram indicate how the force is applied and the distance from the pivot. (2)
- ii. Calculate the moment of the force. (2)
- c. One day, the mechanic hurt his hand and could only exert half the usual force.
 - i. What can he do to produce the same effect? (2)
 - ii. Using a calculation, show that your answer to part (i) is correct. (2)

41. May 2017 Paper I**(10 marks)**

Joseph is teaching his two grandchildren about the use of some tools.

- a. Joseph explain how nails can be removed by using a crowbar. One of his grandchildren gives it a try and applied a force of 15N on the crowbar as shown in the diagram.
 - i. Which **TWO** quantities determine the magnitude of a moment? (2)
 - ii. Calculate the clockwise moment exerted by Joseph's grandchild. (2)
 - iii. Joseph notes that his grandchild is not using the crowbar efficiently. Explain why. (1)
- b. Joseph lifts an object using a rope. Draw arrows to indicate the nature and size of the forces acting on the object being lifted **AND** the direction of the resultant force (if any) when: (5)

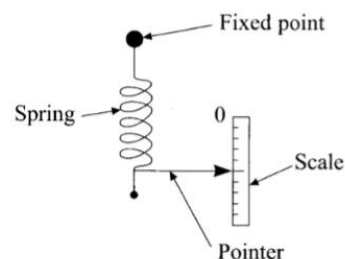
i. The object is stationary;	ii. The object is starting to be lifted;	iii. The object is being lifted at constant speed.

42. September 2017 Paper I

(10 marks)

Luca and Lorraine are investigating the behaviour of a spring.

- a. They suspend it from a fixed point and use a pointer and a vertical scale to find the extension produced when loads are hung from the lower end of the spring. They find that with every 1N of added load the spring extends by 10mm. they make sure that throughout the whole experiment the load does not exceed the elastic limit of the spring.

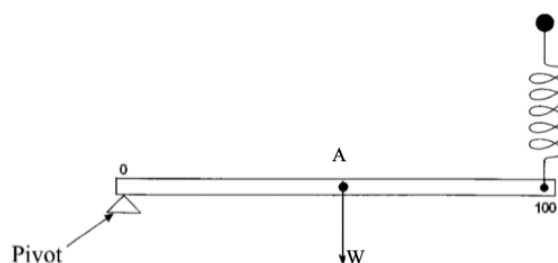


- i. State Hooke's Law. (2)

- ii. State the changes that would occur to the following if the elastic limit is exceeded:

- The extension for every 1N added _____ (1)
- The length of the unloaded spring _____ (1)

- b. Lorraine removes the load from the spring and Luca attaches a uniform meter ruler at the 100cm mark to the spring. The other end of the ruler is placed on a pivot at the 0cm mark. They measure the extension produced using the vertical scale as above.



- i. W represents the weight of the ruler.

What is the name usually given to point A? (1)

- ii. How do we know that in this case point A lies at the midpoint of the ruler (1)

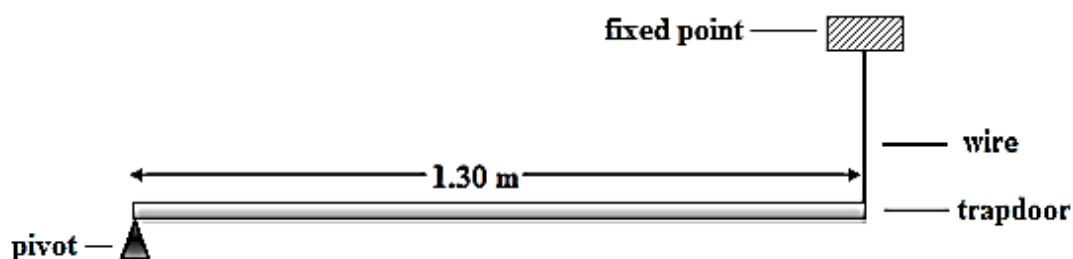
- iii. The extension of the spring produced in this case was measured to be 5mm. calculate the force acting on the spring by the ruler. (Assume that the ruler is horizontal). (1)

- iv. The force just calculated in part (iii) is equal to the upward force on the ruler by the spring. Taking moments about the pivot, calculate the weight of the ruler. (3)

43. April 2016 Paper I**(10 marks)**

Archimedes of Syracuse (287 BC – 212 BC) was a Greek mathematician, astronomer, physicist, inventor and engineer. He is well known for his work on levers. In fact he used to state that if given a lever long enough and a fulcrum on which to place it, he could move the world.

- Why did Archimedes state that the lever has to be long enough? (1)
- Define the term moment of a force. (2)
- State ONE condition for an object to be in equilibrium. (1)
- The diagram below shows a uniform 1.30 m trapdoor of mass 3.00 kg, held horizontally by a wire attached to a fixed point.



- Copy the diagram and on the diagram draw arrows to show TWO forces acting on the trap door apart from the reaction at the pivot. Label these forces. (2)
- Calculate the force that the wire should produce in order to keep the trapdoor horizontal. (2)
- A dog of mass 8.00 kg steps on the trapdoor and sits on it at a distance x metres from the pivot. The trapdoor remains horizontal if the wire produces a force of 76.6 N. Calculate the distance x in metres. (2)

44. September 2015 Paper I**(10 marks)**

A seesaw is 3 m long from end to end.

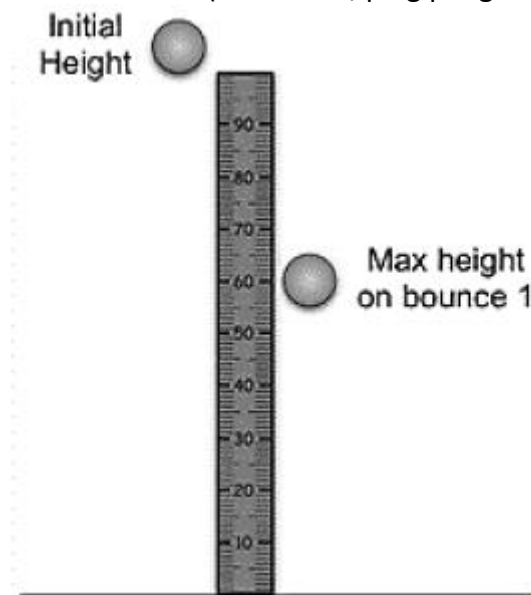
- State the principle of moments. (2)
- If Andrew of mass 35 kg is sitting on one end, how far away from the centre of the see-saw should Daniela, of mass 42 kg, sit to balance the see-saw? (Your answer should include a diagram showing the forces acting on the see-saw.) (3)
- As Andrew is sitting on the balanced seesaw, his cat jumps on him and the see-saw becomes unbalanced. What must Daniela do for the see-saw to balance again? (1)
- Andrew and Daniela go on a seesaw pivoted on a helical spring. When they both sit on opposite sides and the see-saw is balanced, the see-saw is 40 cm above the ground.
 - What happens to the see-saw when both children come off? (2)
 - John and Lisa, each of mass 28 kg sit on the see-saw. Will the see-saw balance again at 40 cm above the ground? Explain. (2)

Energy

45. September 2020 Paper IIA

(20 marks)

Andrea and Melvin want to investigate the relationship between the energy lost on a bounce of different balls. They set up the apparatus as shown in the diagram and decided to use three different balls (tennis ball, ping pong and golf ball).



- a. Copy the diagram below and list the energy changes taking place as the ball is released until it hits the ground. (3)

→+

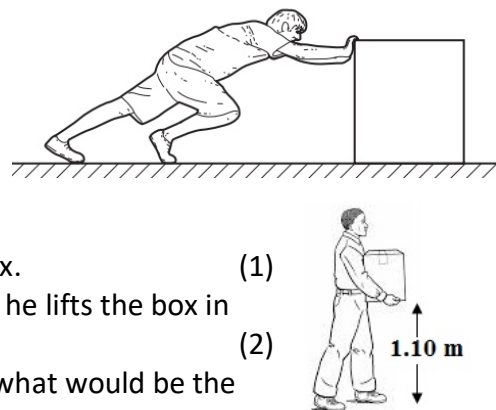
- b. Describe how they should carry out the experiment. (3)
- c. Determine the initial potential energy of the tennis ball if it is released from a height of 1 m and has a mass of 60 g. (3)
- d. Calculate the final velocity of the ball just before it hits the ground. (2)
- e. State the law used to work out the previous question. (2)
- f. After it hits the ground the tennis ball reaches a height of 56 cm. Calculate the potential energy of the tennis ball after it hits the ground. (2)
- g. Calculate the percentage of energy lost after it hits the ground. (2)
- h. Mention **TWO** precautions needed to obtain accurate results. (2)
- i. Energy losses continually occur in everyday life. Describe **ONE** real life situation where a high percentage of energy loss is needed. (1)
-

46. May 2018 Paper I**(10 marks)**

This question is about energy.

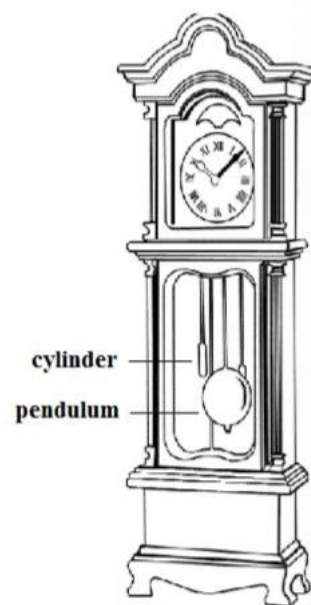
- a. Define the term work. (1)
- b. A man pushes a box a distance of 3.2 m with a constant force of 700 N along a frictionless surface. Calculate the work done by the man. (1)
- c. State the energy changes involved in this process. (1)
- d. The box of mass of 45.0 kg is now lifted vertically upwards by 1.10 m as shown in the diagram.
 - i. Calculate the potential energy gained by the box. (1)
 - ii. What is the power generated by the man when he lifts the box in 3.0 s? (2)
 - iii. If the box had to be released from that height, what would be the kinetic energy with which it would reach the ground? Explain your reasoning by stating the relevant law. (2)
- b. The power obtained is enough to light up a small bulb for a few seconds if it could be converted into electrical energy. State ONE renewable and ONE non-renewable source of electrical energy.

Renewable: _____ Non-Renewable: _____ (2)

**47. May 2017 Paper I****(10 marks)**

A falling object possesses both gravitational potential energy and kinetic energy.

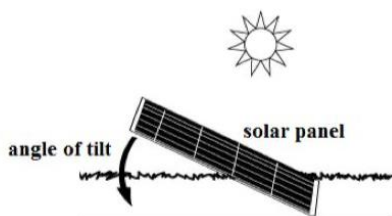
- a. A grandfather clock works by having gravitational potential energy stored in a 4.8kg cylinder. As it descends 1.20m every seven days, this energy is transferred to kinetic energy and the pendulum keeps swinging.
 - i. State the law of conservation of energy. (2)
 - ii. Calculate the gravitational potential energy stored in the cylinder when it is at maximum height. (2)
 - iii. Calculate the power transfer as the cylinder moves downwards. (2)
- b. The transfer of gravitational potential energy into kinetic energy is also used in the hydroelectric process.
 - i. Briefly describe how energy is generated through the hydroelectric process. (3)
 - ii. State **ONE** main advantage of this form of energy. (1)



48. September 2017 Paper I**(10 marks)**

Solar energy is a renewable energy source used to generate electricity.

- c. What is meant by renewable energy source? (1)
- d. Name **TWO** other renewable energy sources used to generate electricity. (2)
- e. A household uses panels of solar cells to generate electricity for this home. The solar panels can be tilted to receive maximum energy input from the Sun.



Month	Angle of tilt		
	15°	30°	55°
January	460	550	430
April	600	630	600
August	650	660	570
December	430	440	410

- i. The table gives the energy received from the sun each second (in J/s) onto a 1m^2 area of solar cells for different angles of tilt during different months of the year.
Use the data in the table to describe how the average energy received by the panel depends on the angle of tilt. (2)
- ii. The total area of the solar cell panels used by the householder is 6m^2 and these are considered to have an efficiency of 22%. Use the data in the table to calculate the maximum electrical energy available from the solar panels each second in August. (4)
- f. Most of the electricity in our homes is still generated from a fossil fuel power station. Give **ONE** advantage of using solar cells to generate electricity compared to using fossil fuels. (1)

49. September 2016 Paper I**(10 marks)**

The Marsa Power station used to depend on the burning of coal. Since it was switched off and put on cold standby in the beginning of 2015 carbon dioxide (CO_2) emissions in the area were reduced by up to 761,000 tonnes a year. (Malta Independent, 21st April 2016)

- a. What is the main difference between renewable and non-renewable sources of energy? (1)
- b. State whether coal is a renewable or non-renewable source of energy. Give a reason for your answer. (2)
- c. Mention **TWO** advantages of using coal as a source of energy. (2)
- d. Mention **TWO** other fossil fuels. (1)
- e. Nuclear energy is one of the alternatives to fossil fuels. State **TWO** disadvantages if Malta had to invest in nuclear energy. (2)
- f. The diagram shows a solar mobile phone charger. On average its solar cells produce 2.7 Joules of energy per second. The solar cells have an efficiency of 15%. Calculate the average input power of the solar cells. (2)

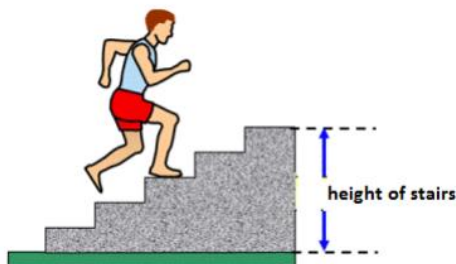


50. September 2016 Paper IIB**(20 marks)**

This question is about personal power.

- a. Work is done when a person goes up the stairs. A number of students investigate how the power generated changes with the weight of a student when going up a set of stairs.

- i. Name the instrument used to measure the: (3)
 1. height of each step:
 2. time for the student to run up the stairs:
 3. weight of the student:



- ii. Use the diagram given to explain how to measure the power of a student running up the stairs. (4)
 - iii. Name TWO precautions taken during this experiment. (2)
 - iv. Would a man with a mass of 60 kg exert a larger force on the stairs than someone with a mass of 90 kg? (1)
 - v. If the two men were to run up the stairs at the same time, explain who will develop more power. (2)
- b. A student weighs 450 N. He runs up the stairs in 6 s. Each step is 20 cm high and there are 30 steps. Calculate:
- i. the total height of the stairs. (1)
 - ii. the work done in climbing the steps. (2)
 - iii. the power the student develops when running up the steps. (2)
 - iv. the potential energy of the student after climbing the stairs. (1)
- c. The student grabs a ball and drops it vertically from the highest step to the ground. Assuming that air resistance is negligible calculate the velocity of the ball as it touches the ground. (2)
-

51. April 2016 Paper I**(10 marks)**

- a. A crane, powered by an electric motor, has a bucket that weighs 460 N when empty. The crane uses the bucket to lift 700 N of concrete up 85 m on a building site.
- Calculate the work done by the crane's motor. (2)
 - Electrical energy in the motor is being converted into other forms of energy as the bucket is lifted upwards (include energy loss). Name these other forms of energy. (2)
- b. A brick falls from the top of the building site and lands vertically 85 m on the ground below in 4.2 s.
- Calculate the final speed of the brick as it hits the ground. (2)
 - Calculate the average speed of the brick whilst it falls to the ground. (2)
 - Describe the energy changes that take place as the brick falls and hits the ground. (2)
-

52. April 2015 Paper I**(10 marks)**

In the first few days of 2015, Malta was hit by a cold storm. People had to heat their homes using either renewable or non-renewable sources of energy.

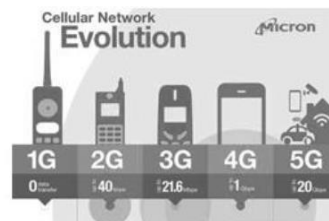
- Explain what a renewable source of energy is. (1)
 - What is a non-renewable source of energy? (1)
 - Give an example of each source of energy (renewable and non-renewable) which may be used for heating homes. (2)
 - Give and explain one **disadvantage** of a renewable source of energy and an **advantage** of a non-renewable source mentioned in part (c). (2)
 - Suggest and explain **one** other action that can be done so that the house remains warmer during the cold months. (2)
 - An electric heater is rated at 1500 W. If the heater is rated as 80 % efficient, calculate the output rate of energy. (2)
-

Waves

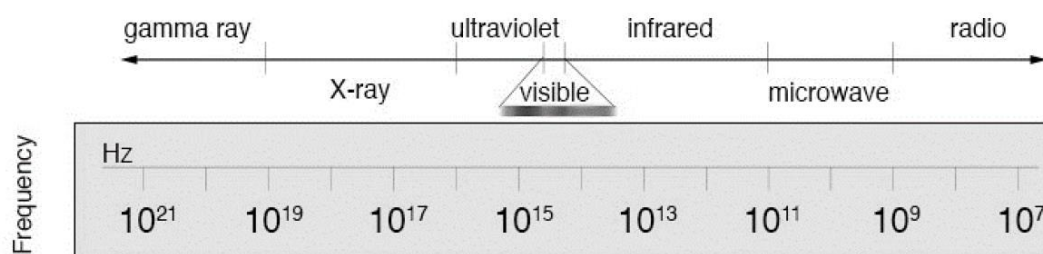
53. September 2020 Paper I

(10 marks)

Wireless technology can use both 4G and 5G for network transmission. Both use a type of electromagnetic wave to transmit energy, but the difference lies in the frequency used. 4G uses frequencies ranging from 1 GHz to 6 GHz while 5G networks use frequencies that can go up to 100 GHz. (1 GHz = 1×10^9 Hz)



- Are electromagnetic waves transverse or longitudinal waves? (1)
- Define the term frequency. (1)
- By looking at the representation below, which part of the electromagnetic spectrum do 5G network use? (1)



- Calculate the wavelength of the 100 GHz, 5G network, if the speed of electromagnetic waves is 3×10^8 m/s. (3)
- Copy the table below and list a use for each of the given electromagnetic wave. (4)

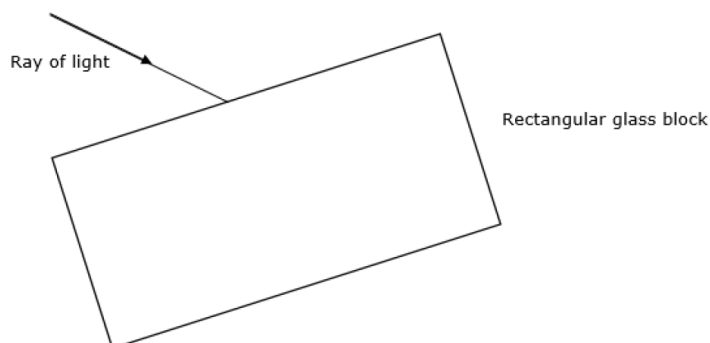
Type of wave	Use
Gamma rays	
X- ray	
Ultraviolet	
infrared	

54. September 2020 Paper IIA**(20 marks)**

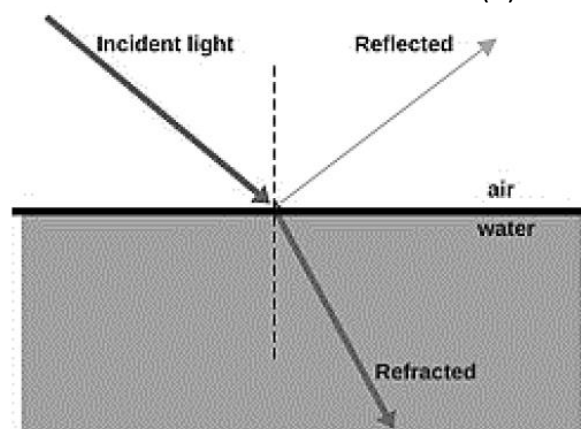
Michele is investigating properties of light as it passes through a rectangular glass.

a. He shines a ray of light onto the rectangular glass block as shown in the figure below.

- i. Copy and complete the diagram to show how the ray of light emerges out of the glass block. The normal lines should be included in the diagram. (3)

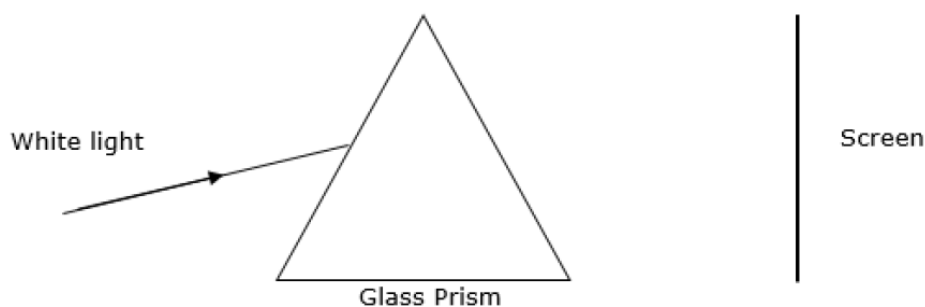


- ii. Mark the angle of incidence (using the letter i) and the angle of refraction (using the letter r). (2)
- iii. Michele finds out that the speed of light in air (3×10^8 m/s) decreases to 2×10^8 m/s when it enters glass. Calculate the refractive index of glass. (2)
- iv. This experiment was repeated using water instead of the rectangular glass block. The outcome obtained will be different, explain what he will observe and why this happens. (2)
- v. At the point where it enters water, some of the light from the ray box seems to be getting reflected rather than refracted through water. Explain why this happens. (1)

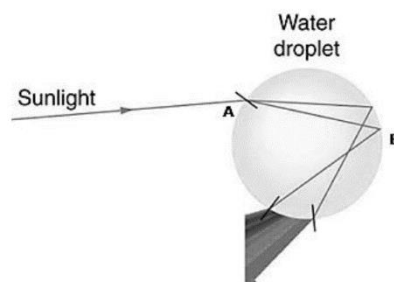


b. He repeats the experiment using a glass prism and a beam of white light.

- i. Copy and complete the diagram below by drawing the path of light through the prism and its projection on the screen. (3)



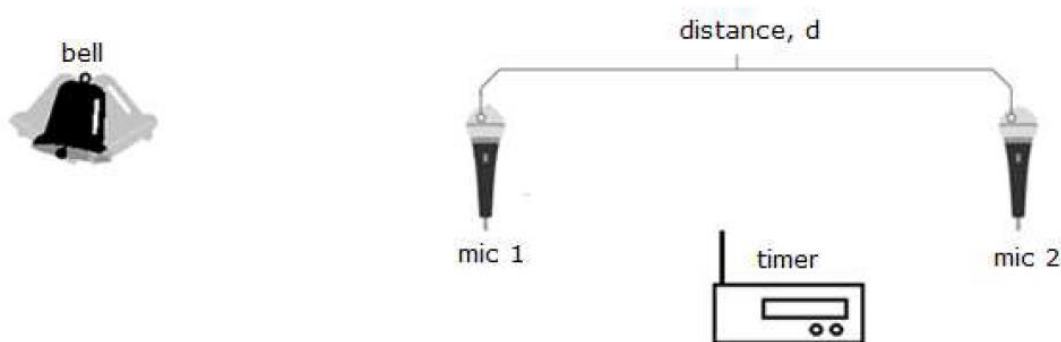
- ii. What is the name of this phenomena? (1)
- iii. Explain why this happens. (2)
- iv. On the screen indicate the colour that will be observed at the top and the one that will be seen at the bottom of the screen. (2)
- v. Rainbows are produced by using the same principle mentioned above. An example is seen in the diagram below. Name the phenomena taking place at A and B. (2)



55. May 2019 Paper I

(10 marks)

An experiment to measure the speed of sound in air was carried out in an open field.



As the bell rings, the sound is received by the two wireless microphones placed at a measured distance, d , apart. The timer records the time taken by the sound wave to travel from microphone 1 to microphone 2.

- a. What type of wave is a sound wave? (1)
- b. Explain how these waves travel through air. (1)
- c. The following set of data was recorded:

Measured distance d/m	15	25	35	45	55	65
Time t/s	0.05	0.08	0.11	0.14	0.17	0.20

- i. Plot a graph of measured distance d/m (y-axis) against time t/s (x-axis). (4)
- ii. Use your graph to calculate the speed of sound in air. (2)
- iii. The experiment is repeated under water, where the microphones can still detect the sound of the bell. State and explain any changes in the time of travel of the wave. (2)

56. May 2019 Paper IIA**(20 marks)**

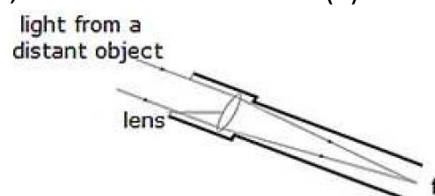
- a. The table below shows different telescopes used to detect radiations from different sources.

Telescope	Type of radiation detected	Wavelength detected
Mauna Kea Observatory (Hawaii)	Infrared	$25 \times 10^{-6} \text{ m}$
Mount Graham (Arizona)	Visible light	$600 \times 10^{-9} \text{ m}$
Effelsberg Observatory (Bonn, Germany)	Radio Waves	0.21 m

- Name **TWO** other properties common to the radiations detected. (2)
- Mention **TWO** other electromagnetic radiations whose wavelengths are smaller than the radiations captured by the telescopes mentioned above. (2)
- Calculate the frequency of the infrared waves detected by the Mauna Kea Observatory. (2)
- State a use of the following radiations: Infrared, Radio waves (2)

- b. The diagram shows part of a telescope.

Light from a distant object is captured by a lens and focused at a point f , as shown.

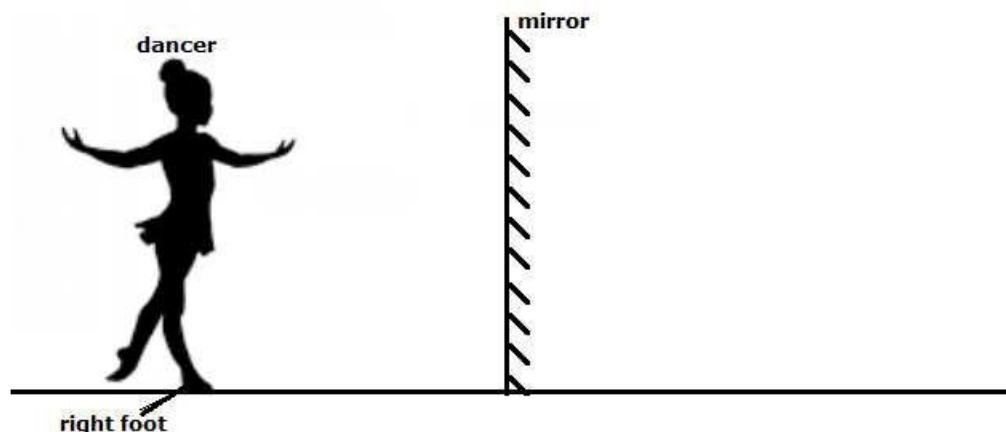


- What type of lens is used in this telescope? (1)
 - When parallel rays pass through the lens, they _____ and meet at the focal point, f . Insert the missing word. (1)
 - The telescope's lens has a focal length of 120 mm. Describe briefly how a student can do a rough verification of the focal length of a lens. (3)
- c. The student wants to use the same lens as a magnifying glass, to be able to read some small text.
- Suggest a distance, at which the lens should be placed away from the text for this to happen. (1)
 - An object whose actual height is 1.5 cm, appears to be enlarged when viewed through the lens. If the magnification of the lens is 4, find the height of the image. (2)
 - Draw a ray diagram to show how the lens produces an image when used as a magnifying glass. Label the object O and the image I . (2)
 - The image produced is enlarged. Mention **TWO** other characteristics of the image formed. (2)

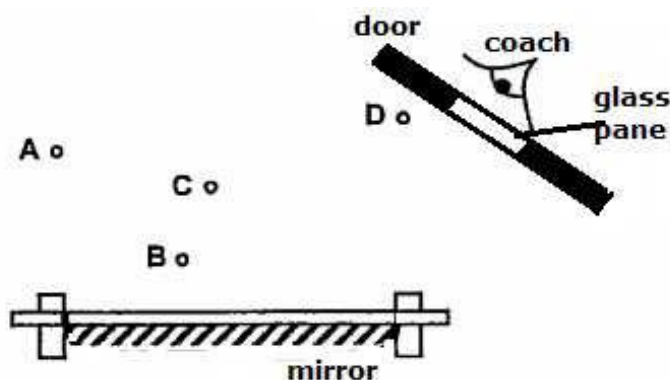
57. September 2019 Paper I**(10 marks)**

Dancers often stand in front of mirrors so that they can see their moves as they practice. The diagram shows a dancer in front of the mirror.

- a. Copy the diagram and draw a ray to show how the dancer can see her right foot through the mirror. Include the path of the ray and the normal. (3)



- b. Mark with a cross (x) the position of the image of the dancer's right foot. (1)
 c. The dancer notices that when she lifts her right foot, the image lifts the left foot. This is because the image is _____. Insert the missing word. (1)
 d. Mention TWO other properties of the image observed in the mirror. (2)
 e. The coach observes four dancers, A,B,C and D, through the glass pane on the door of the dance studio.



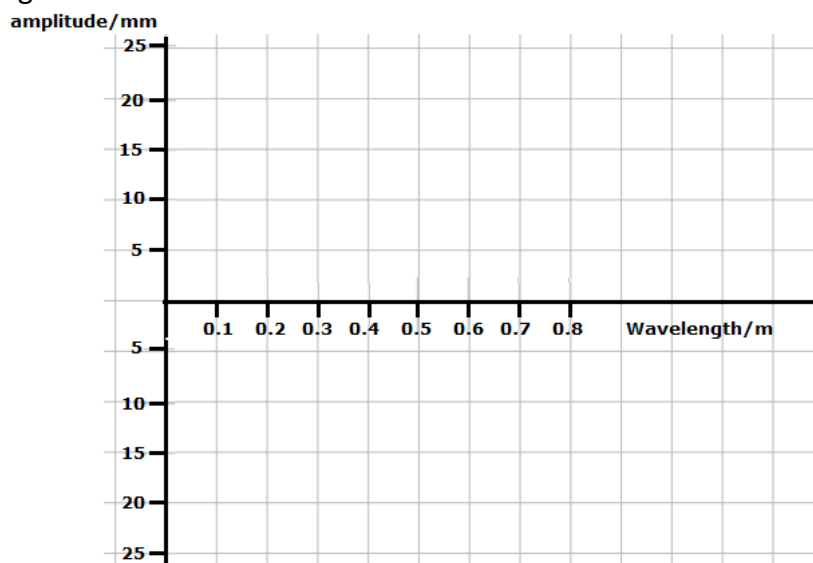
- i. Which dancer cannot be seen by the coach? (1)
 ii. Explain why this dancer cannot be seen by the coach (2)

58. September 2019 Paper I**(10 marks)**

The figure below shows particles on the surface of water. A wave moves from left to right.



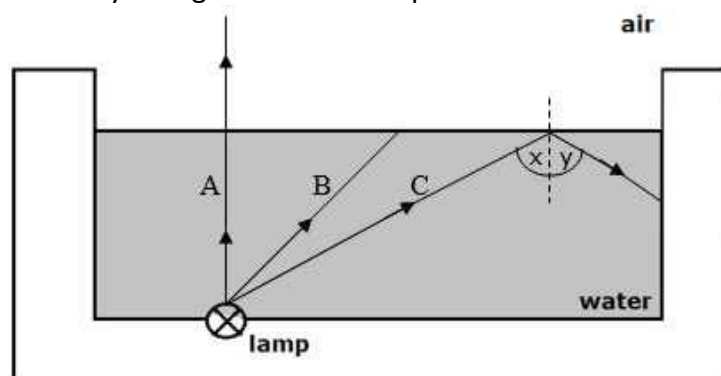
- d. Explain the movement of each particle if the wave is a transverse wave. (2)
- e. Copy the graph below draw a transverse wave that has amplitude of 20 mm and a wavelength of 60 cm. (2)



- f. The wave has a periodic time of 4×10^{-4} s.
- i. Find the frequency of the water wave. (2)
- ii. Calculate the speed of this wave. (2)
- g. The water wave enters a shallower region. Mention **TWO** changes that happen to the water wave as it enters the shallower part. (2)

59. September 2019 Paper I**(10 marks)**

A swimming pool is lit by a lamp at the bottom of the pool, 1.2 m deep. The directions of three rays of light from the lamp are shown.



- Explain why ray A does not bend as it exits the water and travels in air. (1)
- State ONE change that occurs to the light wave as it exits the water. (1)
- Copy the diagram (just for ray B) and continue the path of ray B in air. (1)
- Ray C goes through total internal reflection.
 - Give ONE condition necessary for this phenomenon to take place. (1)
 - What are the names given to angle x and angle y above? (2)
 - What can you say about angle x and angle y? (1)
- When a boy looks down at the lamp, he estimates it to be 90 cm below the surface of the water. Calculate the refractive index of water. (2)

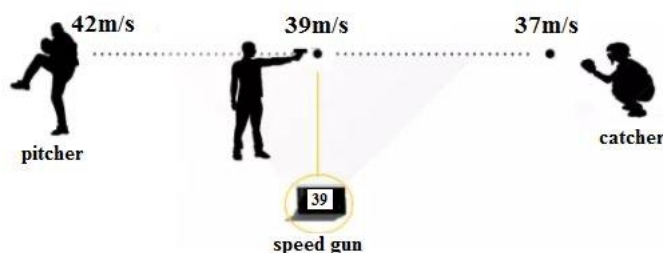
60. May 2018 Paper I**(10 marks)**

The electromagnetic spectrum consists of seven different types of radiation.

- The diagram below shows some of the regions of the electromagnetic spectrum. Complete the diagram by labelling regions A, B and C in the spaces below. (3)

Gamma	A: _____	B: _____	Visible light	C: _____	Micro waves	Radio waves
-------	-------------	-------------	------------------	-------------	----------------	----------------

- State **TWO** common properties of all electromagnetic waves. (2)
- Which of the electromagnetic waves has the lowest frequency? (1)
- State **ONE** main use of gamma radiation. (1)
- Microwaves are used in speed guns which are devices used in sports to detect the speed with which balls are being launched. The diagram below shows one similar example from baseball, where the speed gun monitors the speed of the ball as it travels from the pitcher to the catcher.



- One wavelength of microwaves is 0.0125 m. Calculate the corresponding frequency given that the velocity of infrared rays is 3×10^8 m/s. (2)
- Some of the microwaves transmitted might be absorbed by the ball. What effect would this have on the ball? (1)

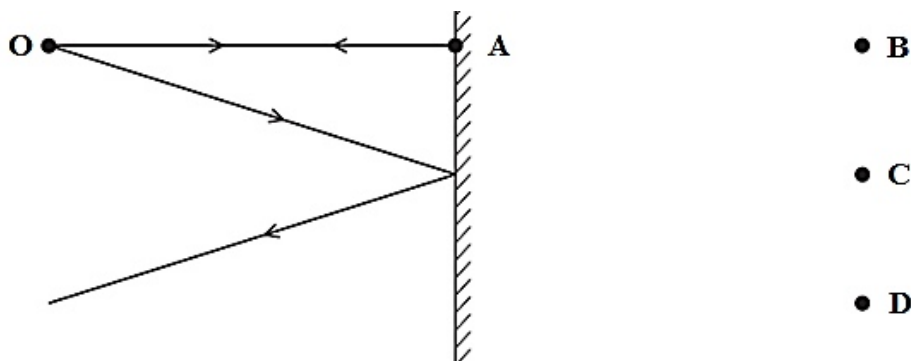
61. May 2018 Paper IIA

(20 marks)

This question is about Optics.

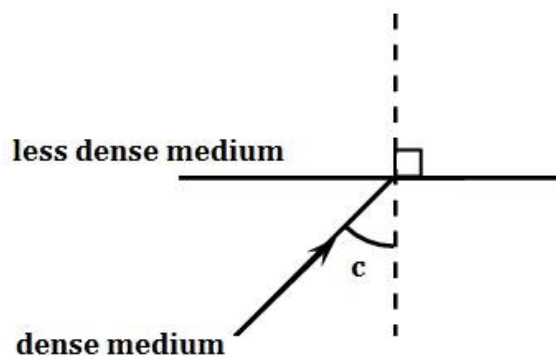
- a. State the law of reflection of waves. (1)
- b. The diagram shows two rays of light from an object O being reflected from a plane mirror.

Complete the diagram to show whether the image will be formed in position B, C or D.



- c. The ray of light shown in the diagram below reaches the boundary between two media at the critical angle.

- i. Complete the diagram to show how the ray will behave at the boundary. (1)
- ii. What will happen to the ray of light if the angle of incidence had to be increased further? (1)
- iii. Give an application in which the situation in c(ii) is used. (1)



- d. An experiment is carried out to investigate refraction of light through a glass block, for angles smaller than the critical angle. You are provided with a ray box connected to a power supply, a semi-circular glass block, plain paper, protractor, pencil and a ruler.
- i. Draw a labelled diagram in the space below to show how you would set up the apparatus for the investigation. The block is already drawn for you. (2)



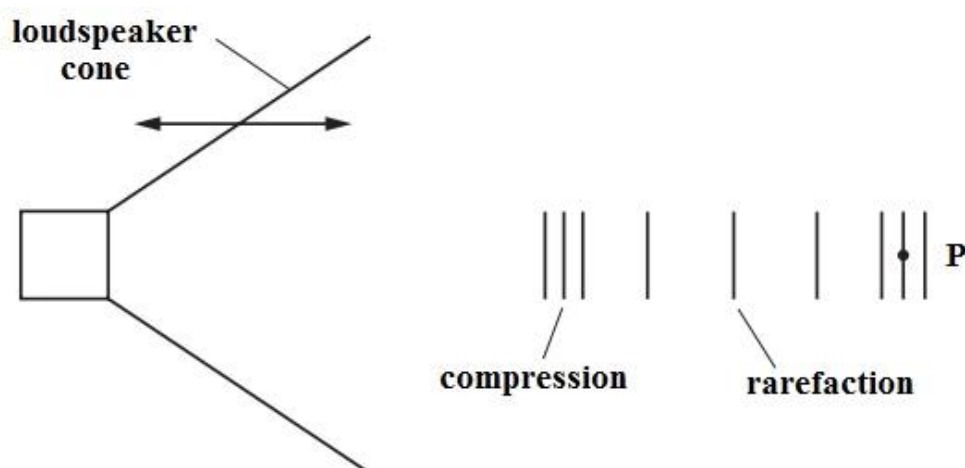
- ii. Give a brief explanation of the method you would use. (3)
- iii. On your diagram, mark clearly the normal at the block and the path the ray of light would follow as it enters and emerges from the block, indicating

- clearly the angle of incidence with an 'i' and the angle of refraction with an 'r'. (3)
- iv. What happens to the speed of light when it enters the block? (1)
- v. The speed of light in air is 3.0×10^8 m/s and the refractive index of the glass block used is 1.495. Calculate the speed of light inside the block. (2)
- vi. Refraction of light occurs also as light passes through lenses to form an image. A converging lens of magnification 3 forms an image 12 cm away from its centre. Calculate the distance in metres between the object and the lens. (3)

62. May 2018 Paper IIA**(20 marks)**

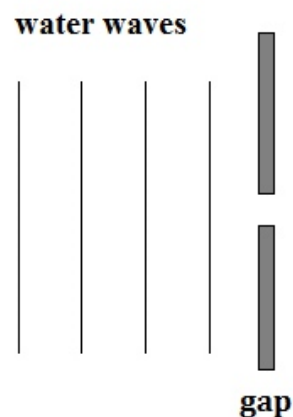
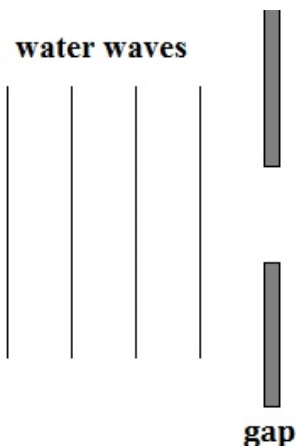
This question is about Waves.

- a. Waves carry energy from one location to another. In the diagram below, compressions and rarefactions are sent out from a loudspeaker cone as it vibrates backwards and forwards with a frequency of 50 Hz.



- i. Is the wave shown in the diagram longitudinal or transverse? Explain your answer in terms of the particles' motion. (2)
- ii. What information does the phrase 'the frequency of vibration is 50 Hz' give us about the vibration? (1)
- iii. P is a compression point. How much time passes before the next rarefaction arrives at P? (2)
- b. Water waves spread out when passing through a gap.
- i. What type of waves are water waves? (1)
- ii. What is the phenomenon observed when a wave passes through a gap called? (1)

- iii. Continue the diagrams below, showing the behaviour of waves as they pass through these two gaps. The diagrams should include the wavefronts and the wave direction. (2)



- c. Water waves are also refracted at a boundary when passing from a deep to a shallow region. State the effect, if any, this has on their:
- frequency _____ (1)
 - wavelength _____ (1)
 - speed _____ (1)
- d. X-rays and Radio waves form part of a group of waves that can travel through vacuum.
- What is the name of this group of waves? (1)
 - List **TWO** other properties that these waves have in common. (2)
 - What is name of the wave with the shortest wavelength from this group of waves? (1)
 - Infrared waves are thermal waves. In fact, we experience this type of heat from a fire or a radiator amongst others. Given that the speed of infrared radiation is $3.0 \times 10^8 \text{ m/s}$ in air and its periodic time is $2.5 \times 10^{-15} \text{ s}$, calculate the wavelength of infrared radiation. (4)
-

63. September 2018 Paper I**(10 marks)**

The electromagnetic (EM) spectrum consists of seven types of radiation, travelling in the form of transverse waves.

- Describe the particle movement in transverse waves. (1)
- State **TWO** properties that are identical for the seven forms of radiation in the E.M. spectrum, apart from their transverse nature. (2)
- Which of the types of radiation in the EM spectrum has the shortest wavelength? (1)
- Give a use for the type of radiation stated in part (c). (1)
- Which of the EM waves can be used to detect fake bank notes? (1)
- TV remote controls are infra red radiation of frequency 3×10^{14} Hz, travelling at 3×10^8 m/s to send signals to a TV set placed 3.50 m away.
- Calculate the number of complete waves in this distance. (3)

**64. September 2018 Paper I****(10 marks)**

Sound is a form of longitudinal wave and it requires a medium in which to travel.

- Complete the following paragraph by writing the correct words that fit in the spaces, to explain how energy is transferred in a longitudinal wave.

___(i)___ vibrate ___(ii)___ to the ___(iii)___ of travel of the wave. This forms a series of ___(iv)___ and ___(v)___ . The distance between these two, is equal to half a complete ___(vi)___ . (3)

- Ultrasound is also a longitudinal wave, with different frequencies than the audible range. State the approximate frequency above which a sound wave is then considered to be an ultrasound wave. (1)

- Bats use ultrasound to form images of their environment by producing waves and then detecting the reflected wave.

Consider a bat at rest, producing ultrasound waves at a frequency of 100 000 Hz, that travel with a speed of 330 m/s.

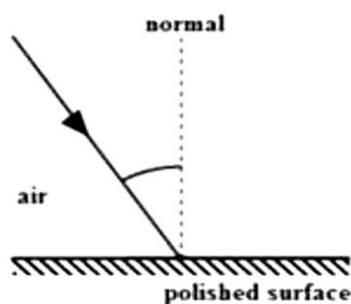
These waves hit a nearby rock and are reflected with practically no changes in frequency. The bat detects them 0.0091 s after emission.

- What does a frequency of 100 000 Hz mean? (1)
- What is the periodic time of this wave? (2)
- How far is the rock from the bat? (2)

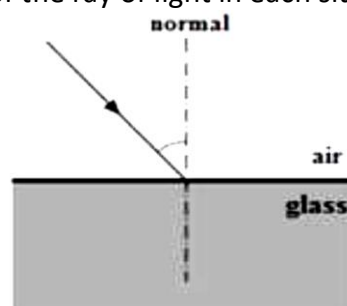


65. September 2018 Paper I**(10 marks)**

- a. A ray of light is travelling in air as shown in the diagrams below. Copy the diagrams and continue them to show the path of the ray of light in each situation.



Situation 1



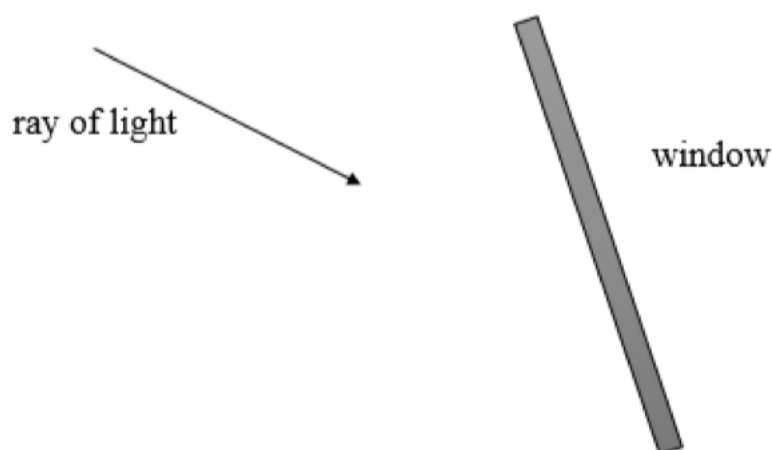
Situation 2

- b. Emma's sunglasses fell into a shallow area of a swimming pool. She inserted her hand into the water to pick them up, but realised that they were at a water level much deeper than she thought.
- When light enters from air into water, its velocity changes. Which **ONE** of frequency and wavelength will change as well? (1)
 - The refractive index of the water in the swimming pool is 1.33. If the sunglasses appear at a depth of 0.080 m, what is the actual depth of the pool? (2)
- c. An object of height 1.5 cm is placed at 3.0 cm from the optical centre of a converging lens of focal length 2.0 cm.
- Draw an accurate ray diagram to scale, to show how the image is produced in this case. Use an arrow to represent the object. Label the object O and the image I. (3)
 - Use accurate measurements to calculate the magnification of the lens. (2)

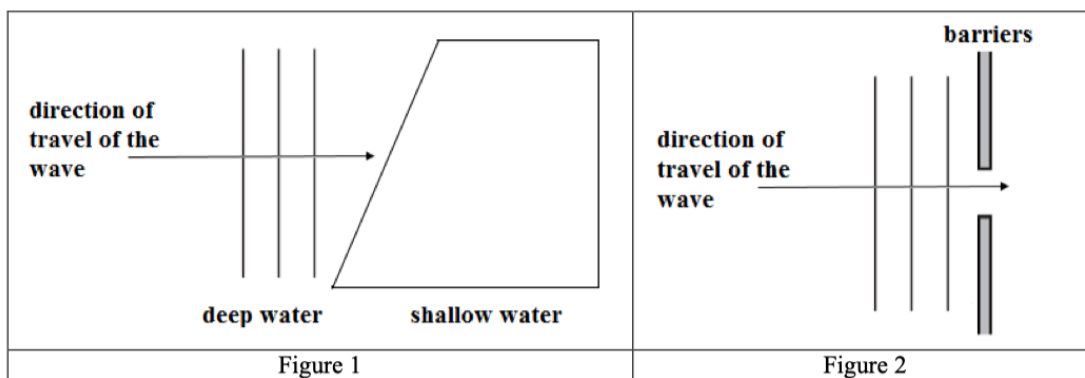
66. May 2017 Paper IIA**(20 marks)**

This question is about waves.

- a. Energy can be transferred from one point to another using transverse or longitudinal waves. Explain how particles move in these two types of waves.
- Transverse: (1)
- Longitudinal: (1)
- b. While playing football, sunlight is reflected from the window of a nearby building into the footballer's eyes.
- Are light waves transverse or longitudinal? (1)
 - State the law of reflection. (1)
 - Complete the diagram below to show the reflection of the ray from the window. Include the normal in your diagram and label clearly the two angles involved. (2)



- iv. What could be done to stop sunlight being reflected into the footballer's eyes from this window? (1)
- c. Light is part of the electromagnetic spectrum.
- State **ONE** factor that all parts of this spectrum have in common, apart from being of the same type of wave mentioned in part b(i). (1)
 - Which part of this spectrum has the highest frequency? (1)
 - Given that this frequency can be as high as $3 \times 10^{19} \text{ Hz}$, calculate the wavelength if the velocity of this wave in air is $2.997 \times 10^8 \text{ m/s}$. (2)
- g. During an investigation about the properties of water waves, the following wavefronts are seen.

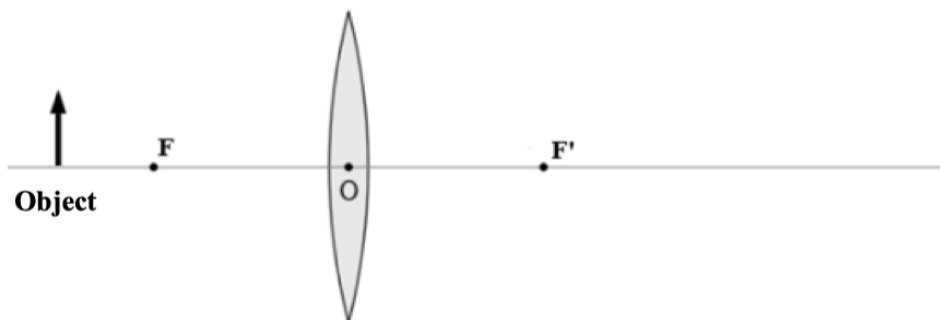


- Complete Figure 1 to show how water waves travel from deep to shallow water, you should include both the wavefronts and the direction of travel of the wave. (3)
- State the name of the phenomenon occurring in Figure 1. (1)
- Complete Figure 2 to show how water travels as it reaches the barriers. You should include both the wavefronts and the direction of travel of the wave. (2)
- State the name of the phenomenon occurring in Figure 2. (1)
- State what happens to the frequency and velocity of the wave when it enters the shallow area in Figure 1. (2)

67. September 2017 Paper I**(10 marks)**

Lenses can be converging or diverging.

- a. The diagram below shows an object, represented by an arrow placed next to a converging lens. F represents the focal point of the lens.
- i. Complete the diagram by using **TWO** rays emerging from the tip of the arrow representing the object, to show the position of the image formed. Illustrate your image using an arrow and label it as 'I'. (3)



- ii. Is the image real or virtual? (1)
- iii. State a practical use for when the object is placed in this position. (1)
- iv. Use a ruler and measure the size of the object and the image, in cm.
Height of object = _____ cm Height of image = _____ cm (2)
- v. Calculate the magnification of the lens used. (1)
- b. As the rays of light enter the convergent lens, their direction will change.
- i. State the name of this phenomenon. (1)
- ii. State the changes, if any, in the direction of the rays if instead of a converging lens, they entered a diverging lens. (1)
-

68. September 2017 Paper IIB**(20 marks)**

This question is about waves.

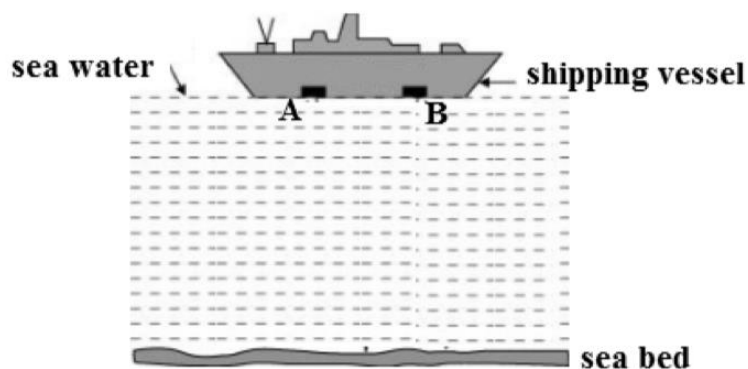
- a. Fill in the missing blanks using the given words.

Longitudinal parallel energy transverse perpendicular

Water waves are _____ (i) _____ waves, since the particles vibrate _____ (ii) _____ to the direction of travel of the wave. Waves carry _____ (iii) _____ in the process. On the other hand, sound waves are _____ (iv) _____ waves and the particles vibrate _____ (v) _____ to the direction of travel of the wave. (5)

- b. A fishing vessel uses ultrasonic pulses to detect the depth of the water. The diagram below shows a vessel with an ultrasound transmitter at point A and a detector at point B.

- ii. Copy the diagram and use lines to show the path of the ultrasound as it is emitted at A and received at B. (2)



- iii. Calculate the time taken for a pulse to travel from point A to point b, if the depth of the water at this instant is 40m. the speed of ultrasound in sea water is 1500m/s. (2)
- iv. Give another use of ultrasonic pulses by fishing vessels. (1)
- v. Circle the correct value above which the longitudinal wave would be considered as ultrasound. (1)
- 20Hz 200Hz 2000Hz 20000Hz
- c. Ocean waves, which travel in the open sea at 5m/s arrive at a beach once every 4s. calculate the distance between the peak of crest and the peak of trough in open sea. (4)
- d. The diagram below shows two musical instruments; the drum and the violin, together with a description of the sound they produce. Underline the correct word (High or Low) to describe the frequency and amplitude of both the drum and the violin. (4)

Musical instrument	Sound produced	Frequency	Amplitude
Drum 	Very loud Low pitch	High OR Low	High OR Low
Violin 	Now very loud High pitch	High OR Low	High OR Low

- e. The diagram below shows a speaker which is switched on close to a human ear. On the diagram, draw the air particles to show their motion as the sound wave travels towards the ear. (1)
-

69. September 2016 Paper I**10 marks**

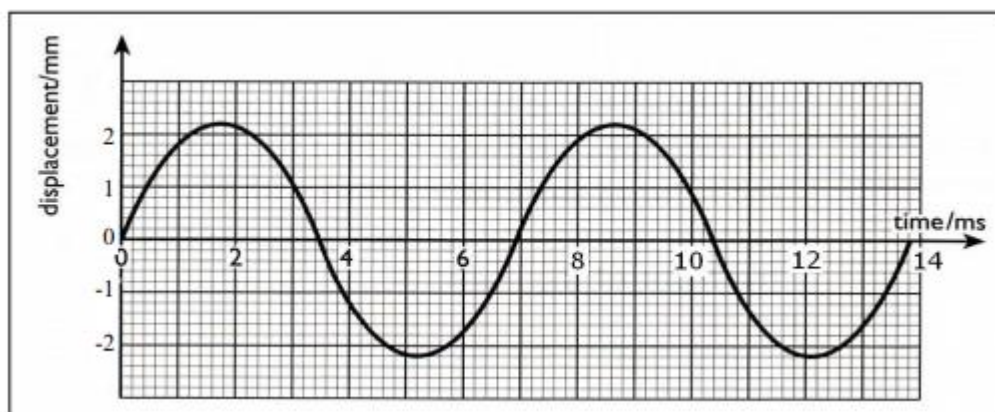
Sir Isaac Newton deduced that the weight of an object is due to the force of gravity. In August 2012, the Curiosity rover landed on Mars. The rover has a mass of 900 kg. The acceleration due to gravity on Mars is 3.7 m/s^2 .

- a. Find:
 - i. The weight of Curiosity on Earth. (2)
 - ii. The mass of Curiosity on Mars. (1)
 - iii. The weight of Curiosity on Mars. (1)
 - iv. Compare the pressure exerted by Curiosity on Mars and Earth. Explain. (2)
- b. The Curiosity rover communicates with Earth using radio waves. Radio waves form part of the Electromagnetic Spectrum.

Name two other parts of the electromagnetic spectrum and give one use for each of them. (4)

70. September 2016 Paper I**(10 marks)**

- a. By means of a diagram, distinguish between a transverse wave and a longitudinal wave. (2)
- b. The graph below represents an object oscillating up and down about a fixed point.

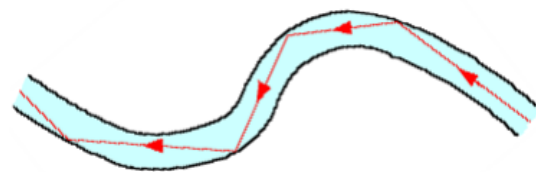


- i. What is the amplitude of the oscillating object? (1)
 - ii. What is the periodic time of the oscillating object? (2)
 - iii. Calculate the frequency of the oscillating object. (2)
 - c. A wave has a wavelength of 25 mm and a frequency of 3 Hz.
 - i. Calculate the velocity of this wave. (2)
 - ii. Show by means of a diagram, what it means to say that a wave 'dies out'. (1)
-

71. September 2016 Paper I**(10 marks)**

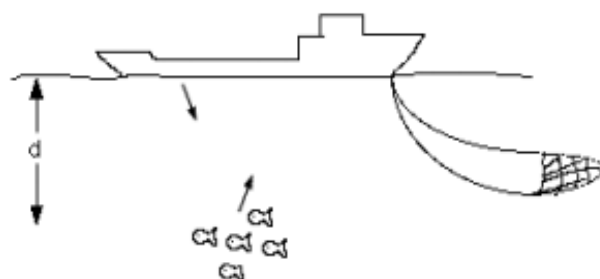
- a. The diagram below shows a light signal travelling through an optical fibre made of solid glass.

- State TWO changes that happened to the light when it passed from air into the glass fibre. (2)
- Explain why light does not escape out of the optical fibre. (2)
- Choose the correct word for the effect shown in the above diagram. (1)

*Total internal reflection**Diffraction**Total external reflection*

- iv. Optic fibres are used locally for internet communications. Name ONE advantage that optic fibres have for this use. (1)

- b. Fishermen use pulses of sound to measure the depth of fish beneath the fishing boat. The sound pulses are reflected from the fish and return to the boat.

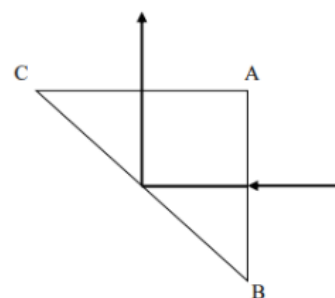


- The speed of sound in sea water is 1500 m/s and the time taken for the sound pulse to return to the boat is 0.2 s. Calculate the depth of the shoal of fish. (2)
- Suggest ONE reason why there may be a second weak returning pulse. (2)

72. April 2016 Paper IIA**(20 marks)**

This question is about light.

- a. Sunlight is made up of different colours and invisible radiations.
- Describe a simple experiment to show the presence of the different colours in light. (Include a diagram). (3)
 - Name TWO radiations in sunlight that the eye cannot detect. (2)
 - Give ONE use for each of these radiations. (2)
- b. A right angled isosceles glass prism is used to deviate a ray of light through 90° as shown in diagram.
- Give TWO reasons why the ray changes direction on side BC, as shown in the diagram. (2)
 - State ONE practical use of a glass prism of this type. (1)
 - Explain why the ray of light passes undeviated from air to glass, on side AB of the prism, as shown in the diagram. (2)
- c. A convex lens has a focal length f , of 40 mm. An object of height 10 mm is placed 30 mm away from the lens.
- Draw a ray diagram (to scale) showing how the image is formed. (4)
 - State whether the image formed is real or virtual. (1)
 - Measure the image distance from the lens. (1)
 - Use your diagram or otherwise, calculate the magnification of the lens. (2)

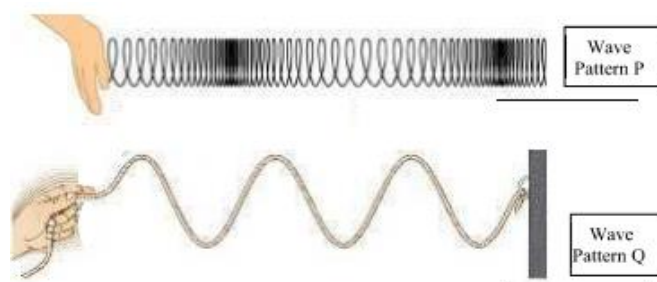


73. April 2015 Paper IIA

(20 marks)

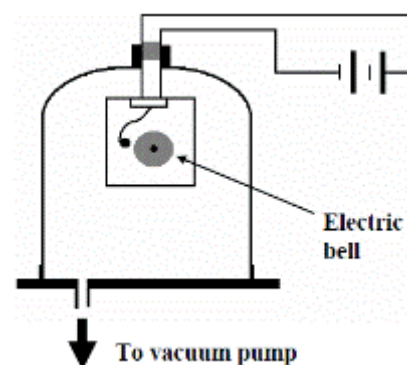
This question is about wave motion

- a. Ms Grech was demonstrating wave motion in the lab using a slinky spring and a length of rope. She obtained the following wave patterns.



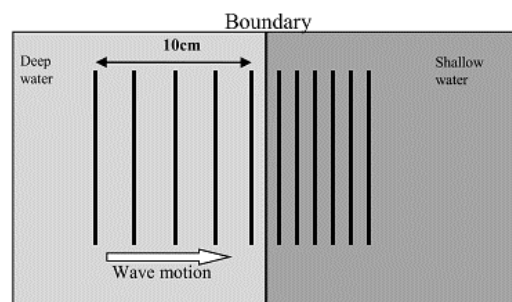
- Name the wave patterns being produced in P and Q. (2)
- What is being transferred when waves go from one side to the other? (1)
- State which pattern represents:
 - A sound wave. (1)
 - An x-ray. (1)

- b. The following lesson, Ms Grech prepared the following apparatus to discuss an important characteristic of sound.



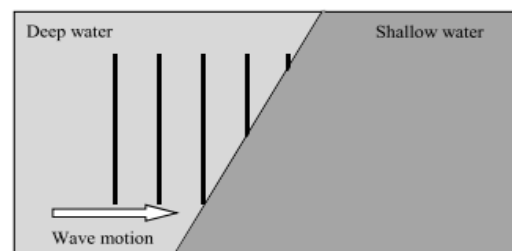
- The electric bell is switched on. Describe two important observations that can be made after the vacuum pump has also been switched on and all the air was removed from the jar. (2)
- What can you conclude from this experiment about:
 - Sound waves (1)
 - Light waves (1)
- Ms Grech hangs the electric bell by using a string. Explain why. (1)

- c. The following diagram represents the motion of water waves, produced by a vibrating long bar, from deep water to shallow water. **(Diagram not to scale)**



- What is the wavelength of the water wave in the deep water? (1)
- The long bar makes 40 vibrations in 8s. Calculate the frequency of the waves produced. (2)
- Calculate the speed of the water wave in the deep water. (3)

- d. The direction of the boundary is now changed as shown.



- Copy and complete the diagram to show the water wave in the shallow water. (2)
- Which phenomenon does the change in direction of the water wave represent? (1)
- What causes this change in direction? (1)

74. April 2015 Paper IIA**(20 marks)**

This question is about pressure and sound waves.

A submarine X is travelling at depth of 64 m below the surface of the sea. The density of seawater is 1050 kg/m^3 and the atmospheric pressure is 100,000 Pa.

a. Calculate:

- i. The pressure on the submarine due to the seawater at a depth of 64 m. (2)
- ii. The total pressure exerted on the submarine at this same depth. (1)
- iii. Two other submarines Y and Z are travelling at a depth of 50 m and 110 m respectively. Compare the pressure exerted by the seawater on submarine X with that of submarines Y and Z. (2)

b. The figure shows the hatch found at the top of the submarine. The hatch has a diameter of 0.85 m. Assuming that the hatch is 64 m below sea level:

- i. calculate the downward force acting on the horizontal hatch. (3)
- ii. state two reasons why the force needed to lift the hatch is different from the value calculated in b (i). (2)



c. The submarine uses ultrasound waves to detect obstacles.

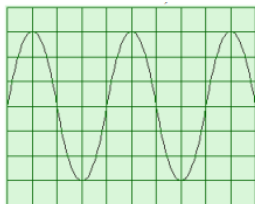
- i. Give two characteristics of ultrasound waves. (2)
- ii. These waves are produced by a vibrating source. Describe how these waves are transmitted through the seawater. (3)

d. The submarine sends an ultrasound pulse which hits an obstacle. This is reflected back and detected by the submarine's receiver. The time taken for the pulse to come back to the submarine is t .

- i. Which other quantity is needed to determine the distance between the obstacle and the submarine? (1)
- ii. How can the distance between the submarine and the obstacle be calculated? (2)
- iii. Give one other use of ultrasound. (1)
- iv. On a particular occasion, the submarine received two different pulses. Suggest a possible explanation for this. (1)

75. September 2015 Paper I**(10 marks)**

A CRO (Cathode Ray Oscilloscope) can be used to show how the voltage changes in a circuit. When a circuit is connected to a CRO, the following image was obtained.



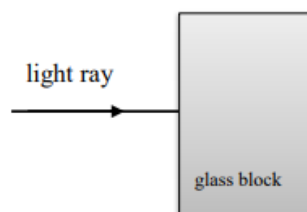
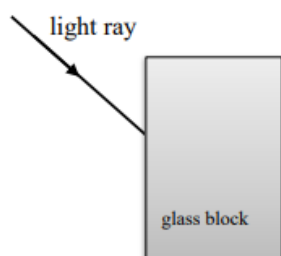
- a. What type of current is flowing in the circuit? Explain. (2)
- b. If the above trace was observed in 0.05 s. Find the frequency. (2)
- c. A diode is connected to a circuit in series to the power supply. Copy the above diagram and draw the new image on the same axis to show what would be seen on the CRO now. (2)
- d. What do we call this process? (1)

- e. Draw in the space provided, the diagram that would be seen on the CRO if the source of the voltage is replaced with a battery. Explain your reasoning. (2)
- f. Name a possible source of voltage in part (a). (1)
-

76. September 2015 Paper I**(10 marks)**

Willebrord Snellius was a Dutch astronomer who studied refraction of light waves in the 17th Century.

- a. Copy and complete the following diagrams. (3)



- b. Name two changes that occur when light is refracted. (2)
- c. A fisherman sees a fish at an apparent depth below the surface of the water of 175 cm. Given that the refractive index of water is 1.33, what is the true depth, in metres, of the fish below the water's surface? (2)
- d. Swimming pool owners fix depth markers to indicate the depth of the pool. With the aid of a ray diagram, explain why a pool filled with water looks shallower. (3)
-

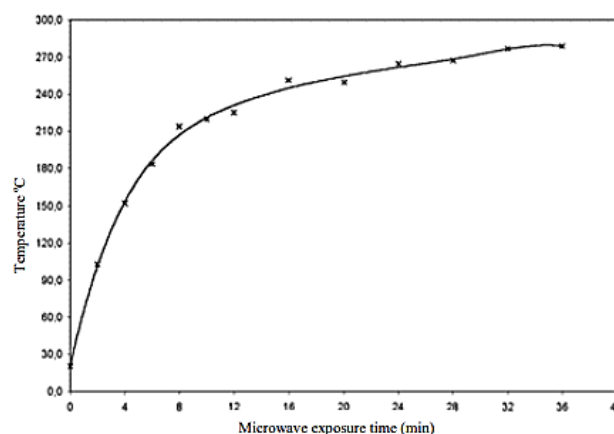


air
water

77. September 2015 Paper I**(10 marks)**

Microwaves are classified as electromagnetic radiation. They are reflected from metals but can pass through glass and some plastics.

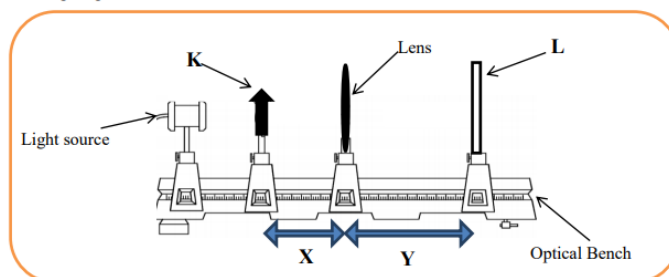
- a. How does food heat up in a microwave? (2)
- b. Name a type of electromagnetic wave which has a longer wavelength and another which has a shorter wavelength than microwaves. (2)
- c. A microwave oven releases microwaves of frequency 4000 MHz. If the speed of light is 3×10^8 m/s, calculate the wavelength of the microwaves. (2)
- d. X-rays and gamma rays are also electromagnetic waves. State one way in which they differ from microwave radiation and one way in which they are similar. (2)
- e. The following is a graph showing the change in temperature with time when heating oil samples in a microwave oven. What is the average change in temperature per **second** during the first 6 minutes?
-



78. September 2015 Paper IIB**(20 marks)**

Joseph and Victoria were on their way home after watching a film at the cinema. They were wondering how a very small projector can produce and project such a very large image onto the cinema screen.

- What do we call this phenomenon? (1)
- What type of lens is needed to produce such an image? (1)
- During their Physics lesson, Joseph and Victoria were supplied with the apparatus shown in the following diagram



- Which of items K and L are:
 - A screen
 - An object
 - What are the distances X and Y called?
- Describe what happens, in terms of light rays, when the light source is switched on.
 - What can Joseph and Victoria do to obtain different values of X and Y while keeping the lens in a fixed place?
 - Give one precaution they had to take.
 - The figure on the right shows scale drawings of a window frame and the image of the frame produced on a screen by a converging lens.
 - Calculate the magnification of the image.
 - The image of the frame was produced 45 mm from the lens. Determine the distance of the actual frame from the lens.
 - Victoria told Joseph that they need to make one change in the window frame setup to be similar to that used in the cinema projector.
 - Explain what needs to be changed.
 - Describe how, using the same apparatus, Joseph and Victoria can find the focal length of the lens.

